# Water Quality Sampling Data Sheet:

Date: 1/12/14 +4 ext
Fishpond:

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 8a 1/12 | % Cloud Cover | 10% very sunny |
| **Finish** Time | 1 night 10a 1/14 | | |
| Tide | low | Precipitation | 0 in last 24 hrs |
| Air Temp | [illegible] | Other Comments: | |
| Wind Speed/Direction | 0 mph 8m | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | | where's | | the | stick? | | | | | |
| Surf. Water Temp °C | 24.76 | 23.09 | 25.06 | 24.33 | 24.03 | 23.56 | 22.00 | 21.79 | 22.30 | 22.62 |
| Surf. Salinity | 33.16 | 32.54 | 31.18 | 31.29 | 32.14 | 32.21 | 28.71 | 28.20 | 31.01 | 32.11 |
| Surf. DO % | 29.4 | 58.8 | 36.2 | [illegible] 60.2 | 60.1 | 58.9 | 77.1 | 76.5 | 71.0 | 52.4 |
| Surf. DO mg/L | 2.04 | 4.17 | 2.50 | 4.16 | 4.19 | 4.15 | 5.65 | 5.63 | 5.27 | 3.77 |
| Surf. pH | 7.63 | 7.81 | 7.59 | 7.77 | 7.79 | 7.76 | 7.86 | 7.82 | 7.82 | 7.77 |
| Bottom Temp °C | 24.32 | 23.50 | 25.22 | 25.17 | 23.89 | 23.54 | 23.56 | 23.17 | 22.77 | |
| Bottom Salinity | 33.16 | 33.26 | 32.50 | 32.35 | 32.52 | 32.71 | 32.36 | 33.56 | 31.89 | |
| Bottom DO% | 25.2 | 58.1 | 34.3 | 45.3 | 52.5 | 56.5 | 69.7 | 71.3 | 71.1 | |
| Bottom DO mg/L | 1.71 | 4.01 | 2.34 | 3.08 | 3.66 | 3.99 | 4.90 | 5.00 | 5.09 | |
| Bottom pH | 7.62 | 7.77 | 7.62 | 7.69 | 7.77 | 7.79 | 7.84 | 7.88 | 7.83 | |
| $NO_3^-$ (Nitrate) mg/L | 0.67 | 0.73 | 0.69 | 0.38 | 0.16 | 0.73 | 0.67 | 1.17 | 1.05 | 1.12 |
| $NO_2^-$ (Nitrite) mg/L | 0.004 | 0.000 | 0.000 | 0.001 | 0.000 | 0.007 | 0.000 | 0.007 | 0.003 | 0.015 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.05 | 0.17 | 0.41 | 0.14 | 0.10 | 0.03 | 0.14 | 0.03 | 0.03 | 0.09 |
| $SiO_2$ (Silica) mg/L | 3 | 4 | 4 | 5 | 4 | 3 | 6 | 4 | 6 | 4 |
| Turbidity (FAU) | [illegible]9 | [illegible]4 | 2 | 0 | 0 | 7 | 4 | 0 | 10 | 4 |

YSI Not working

# Water Quality Sampling Data Sheet:

Date: 1/13/14
Fishpond: Heeia gathered, finished 12:45p

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 11:40a | % Cloud Cover | 10% |
| **Finish** Time | ~~3:15 PM~~ | | |
| Tide | Low | Precipitation | 0 |
| Air Temp | | Other Comments: | low tide<br>Koa winds very light |
| Wind Speed/Direction | 0 | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | | | | | | | | | | | |
| Surf. Water Temp °C | 24.53 | 25.30 | 24.74 | 23.41 | 24.98 | 24:13 | 23.58 | 24.48 | 23.33 | ~~23.44~~ | 25.09 |
| Surf. Salinity | 32.36 | 32.04 | 29.25 | 22.42 | 30.94 | 32.10 | 26.66 | 31.30 | 31.13 | ~~32.28~~ | 32.23 |
| Surf. DO % | 69.2 | 68.6 | 69.9 | 80.0 | 69.4 | 78.6 | 87.1 | 100.0 | 97.9 | ~~100.9~~ | 83.4 |
| Surf. DO mg/L | 4.79 | 4.73 | 4.92 | 5.98 | 4.83 | 5.49 | 6.34 | 6.98 | 6.99 | ~~7.04~~ | 5.72 |
| Surf. pH | 7.78 | 7.79 | 7.80 | 7.68 | 7.83 | 7.86 | 7.78 | 7.93 | 7.97 | 7.95 | |
| Bottom Temp °C | 24.71 | 25.17 | 25.16 | 24.56 | 25.27 | 24.28 | 24.24 | 24.50 | 23.44 | 25.23 | |
| Bottom Salinity | 32.17 | 32.11 | 31.13 | 29.50 | 31.86 | 32.09 | 28.86 | 31.42 | 31.28 | 32.15 | |
| Bottom DO% | 63.3 | 72.8 | 73.1 | 85.1 | 73.9 | 81.6 | 84.2 | 100.7 | 101.9 | 83.3 | |
| Bottom DO mg/L | 4.38 | 4.99 | 5.04 | 6.01 | 5.07 | 5.69 | 5.98 | 7.00 | 7.24 | 5.72 | |
| Bottom pH | 7.86 | 7.84 | 7.83 | 7.87 | 7.84 | 7.89 | 7.83 | 7.95 | 7.97 | 7.86 | |
| $NO_3^-$ (Nitrate) mg/L | 1.001 | 0.009 | 0.006 | 0.008 | 0.000 | 0.006 | 0.000 | 0.002 | 0.003 | 0.010 | |
| $NO_2^-$ (Nitrite) mg/L | 0.28 | 0.64 | 0.49 | 0.69 | 0.33 | 0.11 | 0.34 | 0.78 | 0.78 | 0.30 | |
| $NH_4$ (Ammonia) mg/L | 0.01 | 0.00 | 0.03 | 0.03 | 0.00 | 0.00 | 0.04 | 0.04 | 0.00 | 0.00 | |
| $PO_4^{-3}$ (Phosphate) | 1.25 | 1.27 | 0.65 | 0.73 | 0.26 | 0.36 | 0.22 | 1.33 | 0.32 | 1.04 | |
| $SiO_2$ (Silica) mg/L | 3 | 2 | 6 | 8 | 4 | 3 | 5 | 3 | 3 | 3 | |
| Turbidity (FAU) | 0 | 0 | 7 | 1 | 1 | 0 | 0 | 0 | 0 | 5 | |

Nitrite
Nitrate
Silica
Phos
Amm
Turb

this # is puzzling. it was lighter in color than #10 yet has a higher value. And normally darker color = higher value

# Water Quality Sampling Data Sheet:

Date: 1/20/15
Fishpond: Heeia
Initial Observations: Murky

| | | | |
|---|---|---|---|
| **Start** Time | 10:45 | % Cloud Cover | 25% very clear |
| **Finish** Time | | | |
| Tide | low | Precipitation | 0 |
| Air Temp | 80°F | Other Comments: | • consistent winds<br>• slow rising tide |
| Wind Speed/Direction | 0-3 Onshore | | |

• consistent wind

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 41 | 55 | 41 | 43 | 82 | 46 | 37 | 38 | 48 | 51 |
| Surf. Water Temp °C | 25.55 | 25.67 | 24.24 | 26.51 | 25.71 | 25.83 | 27.69 | 27.43 | 26.62 | 26.59 |
| Surf. Salinity | 34.86 | 33.5 | 26.8 | 34.03 | 33.85 | 34.53 | 34.74 | 34.66 | 34.51 | 34.8 |
| Surf. DO % | 66.3 | 71.7 | 50.8 | 66.0 | 60.8 | 79 | 77.6 | 91.0 | 89.2 | 86 |
| Surf. DO mg/L | 4.45 | 4.81 | 3.80 | 4.37 | 4.08 | 4.87 | 5.03 | 5.84 | 5.89 | 5.68 |
| Surf. pH | 7.81 | 7.83 | 7.55 | 7.80 | 7.78 | 7.83 | 7.85 | 7.87 | 7.89 | 7.88 |
| Bottom Temp °C | 25.57 | 25.60 | 26.37 | 27.06 | 2561 | 25.85 | 27.70 | 27.47 | 26.72 | 26.59 |
| Bottom Salinity | 34.84 | 34.91 | 34.78 | 34.66 | 35.01 | 34.77 | 34.83 | 34.66 | 34.53 | 34.79 |
| Bottom DO% | 65.4 | 73.5 | 54.1 | 71.5 | 66.3 | 75.0 | 81.5 | 89.3 | 86.3 | 85.0 |
| Bottom DO mg/L | 4.42 | 4.93 | 3.61 | 4.56 | 4.36 | 4.90 | 5.27 | 5.78 | 5.74 | 5.62 |
| Bottom pH | 7.82 | 7.84 | 7.65 | 7.83 | 7.83 | 7.85 | 7.85 | 7.87 | 7.89 | 7.89 |
| $NO_3^-$ (Nitrate) mg/L | 0.05 | 0.03 | 0.08 | 0.02 | 0.03 | 0 | 0.03 | 0.03 | 0.04 | 0.03 |
| $NO_2^-$ (Nitrite) mg/L | 0.004 | 0.007 | 0.003 | 0.002 | 0.004 | 0.000 | 0.003 | 0.007 | 0.005 | 0.008 |
| $NH_4$ (Ammonia) mg/L | 0.91 | 1.02 | 0.00 | 1.09 | 0.60 | 1.00 | 1.45 | 1.14 | 1.11 | 0.76 |
| $PO_4^{-3}$ (Phosphate) | 0.05 | 0.11 | 0.13 | 0.12 | 0.08 | 0.13 | 0.10 | 0.29 | 0.05 | 0.02 |
| $SiO_2$ (Silica) mg/L | 2 | 5 | 15 | 1 | 5 | 3 | 3 | 2 | 2 | 2 |
| Turbidity (FAU) | 6 | 4 | 18 | 9 | 13 | 6 | 13 | 6 | 4 | 16 |

# Water Quality Sampling Data Sheet:

Date: 1/22/15
Fishpond: He'eia
Initial Observations: Maury

| | | | |
|---|---|---|---|
| **Start** Time | 8:21 | % Cloud Cover | 35% Voggy |
| **Finish** Time | | | |
| Tide | low dropping | Precipitation | 0 |
| Air Temp | 75°F | Other Comments: | • tide was some level as cylinders @ break flowing out of pond |
| Wind Speed/Direction | 0-5 onshore | | |

• not hot like yesterday but still sunny
• consistent wind

6 3

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | Forgot ruler | on shore | sorry | | | | | | | |
| Surf. Water Temp °C | 24.12 | 23.98 | 24.25 | 25.03 | 24.43 | 25.04 | 24.72 | 24.55 | 24.40 | 24.21 |
| Surf. Salinity | 34.98 | 34.85 | 34.93 | 34.66 | 34.60 | 34.62 | 34.39 | 34.19 | 33.88 | 34.99 |
| Surf. DO % | 60.7 | 57.7 | 61.5 | 45.9 | 51.7 | 41.2 | 62.6 | 66.5 | 62.8 | 63.0 |
| Surf. DO mg/L | 4.18 | 3.98 | 4.22 | 3.09 | 3.56 | 2.70 | 4.26 | 4.54 | 4.33 | 4.29 |
| Surf. pH | 7.76 | 7.76 | 7.84 | 7.74 | 7.79 | 7.67 | 7.83 | 7.86 | 7.82 | 7.85 |
| Bottom Temp °C | 24.13 | 24.16 | 24.27 | 25.01 | 24.61 | 24.97 | 25.04 | 24.82 | 24.41 | 24.24 |
| Bottom Salinity | 35.01 | 35.09 | 35.13 | 34.98 | 35.18 | 34.82 | 34.86 | 35.15 | 34.09 | 34.98 |
| Bottom DO% | 60.1 | 61.8 | 70.8 | 54.6 | 61.5 | 40.8 | 61.0 | 70.4 | 63.2 | 57.4 |
| Bottom DO mg/L | 4.12 | 4.24 | 4.85 | 3.74 | 4.16 | 2.78 | 4.08 | 4.77 | 4.35 | 3.87 |
| Bottom pH | 7.78 | 7.82 | 7.90 | 7.80 | 7.87 | 7.72 | 7.86 | 7.90 | 7.86 | 7.85 |
| $NO_3^-$ (Nitrate) mg/L | 1.33 | 0.84 | 0.96 | 1.18 | 1.30 | 1.22 | 0.95 | 1.00 | 0.92 | 0.85 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.007 | 0.003 | 0.007 | 0.004 | 0.004 | 0.000 | 0.003 | 0.002 |
| $NH_4$ (Ammonia) mg/L | 0.01 | 0.02 | 0.02 | 0.02 | 0.04 | 0.05 | 0.01 | 0.00 | 0.02 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.17 | 0.14 | 0.06 | 0.09 | 0.19 | 0.11 | 0.08 | 0.20 | 0.08 | 0.10 |
| $SiO_2$ (Silica) mg/L | 3 | 3 | 4 | 4 | 3 | 4 | 2 | 3 | 3 | 2 |
| Turbidity (FAU) | 5 | 4 | 0 | 5 | 3 | 4 | 0 | 0 | 3 | 1 |

# Water Quality Sampling Data Sheet:

Date: 1/26/15
Fishpond: He'eia
Initial Observations: Maury

| | | | |
|---|---|---|---|
| **Start** Time | 8:10 am | % Cloud Cover | 100% |
| **Finish** Time | 1:30 pm | | |
| Tide | medium dropping | Precipitation | 0 |
| Air Temp | 72° | Other Comments: | •Consistant & gusty winds<br>•tide dropping however @ the break the wind & tide were appearing to be at a standstill. |
| Wind Speed/Direction | Onshore 10-20 | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 57 | 60 | 56 | 65 | 85 | 53 | 49 | 54 | 66 | 56 |
| Surf. Water Temp °C | 20.46 | 20.14 | 20.13 | 20,31 | 20.85 | 21.47 | 21.38 | 21.56 | 21.52 | 21.04 |
| Surf. Salinity | 18.33 | 10.84 | 3.46 | 11,27 | 23,87 | 27,10 | 25.74 | 26.88 | 28,95 | 25,89 |
| Surf. DO % | 80.3 | 82,2 | 67,8 | 83.6 | 81,4 | 78.5 | 77,3 | 79,2 | 83.9 | 84.0 |
| Surf. DO mg/L | 6,45 | 7.00 | 6,02 | 6.74 | 6,35 | 5,91 | 5,88 | 5,97 | 6,23 | 6,41 |
| Surf. pH | 7,59 | 7,60 | 7,62 | 7,57 | 7,76 | 7,73 | 7,72 | 7,74 | 7,80 | 7,82 |
| Bottom Temp °C | 21,83 | 21,76 | 21.98 | 21,61 | 21,45 | 21,61 | 21,33 | 21,96 | 21,58 | 21,24 |
| Bottom Salinity | 26,98 | 28,42 | 27,16 | 26,41 | 27,29 | 27,46 | 25,01 | 28,70 | 29,70 | 26,40 |
| Bottom DO% | 50.8 | 73.0 | 51,7 | 62,7 | 77,1 | 76,3 | 77,3 | 76,4 | 84,8 | 81,4 |
| Bottom DO mg/L | 3,72 | 5,38 | 3,81 | 4,75 | 5,83 | 5,73 | 5,94 | 5,70 | 6,25 | 6,19 |
| Bottom pH | 7,61 | 7,70 | 7,45 | 7,68 | 7,74 | 7,74 | 7,73 | 7,78 | 7,82 | 7,80 |
| $NO_3^-$ (Nitrate) mg/L | 0,83 | 0,40 | 0,37 | 0,75 | 0,87 | 0,84 | 0,64 | 0,90 | 0,18 | 0,20 |
| $NO_2^-$ (Nitrite) mg/L | 0,007 | 0,002 | 0,001 | 0,004 | 0,005 | 0,005 | 0,000 | 0,011 | 0,003 | 0,011 |
| $NH_4$ (Ammonia) mg/L | 0,01 | 0,03 | 0,02 | 0,00 | 0,02 | 0,02 | 0,04 | 0,02 | 0,03 | 0,00 |
| $PO_4^{-3}$ (Phosphate) | 0,11 | 0,11 | 0,43 | 0,18 | 0,09 | 0,15 | 0,18 | 0,07 | 0,09 | 0,23 |
| $SiO_2$ (Silica) mg/L | 10 | 12 | 21 | 15 | 5 | 6 | 6 | 5 | 3 | 6 |
| Turbidity (FAU) | 7 | 11 | 17 | 16 | 13 | 5 | 12 | 5 | 6 | 8 |

0,002 0,001 0,004 0,005 0,005 0,011 0,003 0,011

19 25.30

21

19
20
39

# Water Quality Sampling Data Sheet:

Date: 4/28/15
Fishpond: Heleia
Initial Observations: Maury

| | | | |
|---|---|---|---|
| **Start** Time | 6:52 pm | % Cloud Cover | 40% |
| **Finish** Time | 7:00pm 1/30 | | |
| Tide | low | Precipitation | 0 |
| Air Temp | 76.6°F | Other Comments: | very calm, became windy later (sites 7→10) clouds cleared as well. very clear |
| Wind Speed/Direction | 0 | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 33 | 42 | ~~27~~ 40 | 34 | 59 | 37 | 35 | 35 | 71 | 43 |
| Surf. Water Temp °C | 27,39 | 26.15 | 26,51 | ~~29~~ 27,12 | 27.75 | 26,65 | 25,73 | 24.01 | ~~29~~ 24.21 | 24,17 |
| Surf. Salinity | 19,53 | ~~25~~ 30,75 | 25.32 | 29,91 | ~~3~~ 31,34 | 27,06 | 18,85 | 17,84 | ~~28~~ 28,73 | ~~1~~ 16.96 |
| Surf. DO % | 78,1 | 97,3 | 73,0 | 104,0 | 108,3 | 96,1 | ~~72~~ 72.8 | 94,8 | 93,01 | 80,5 |
| Surf. DO mg/L | 5,53 | 6.82 | 5,01 | 6,98 | 7,09 | 6,62 | 5,23 | 7,19 | 6,84 | 6,06 |
| Surf. pH | 7,77 | 8,03 | 7,82 | 7.96 | 8,03 | 8,01 | 7,88 | 8,00 | 7,99 | 7,96 |
| Bottom Temp °C | 28,01 | 27,68 | 26,90 | 27,31 | 27,45 | 27,03 | 26,36 | ~~26~~ 25,59 | 24,32 | 26,1 |
| Bottom Salinity | 25,50 | 30,15 | 37,45 | 31,56 | 32,69 | 28,81 | 27,54 | 24,05 | 29,86 | 26,59 |
| Bottom DO% | 68,5 | ~~103~~ 103.6 | 60,7 | 96,0 | 93,0 | ~~3~~ 98,4 | 81,2 | 90,5 | 94,8 | 92,9 |
| Bottom DO mg/L | 4,64 | ~~6~~ 7,00 | 4,02 | 6,32 | 6,03 | 6,60 | 5,63 | 6,34 | 6,69 | 6,39 |
| Bottom pH | 7,81 | 7,98 | 7,73 | 7,92 | 7,95 | 8,01 | 7,89 | 7,97 | 7,98 | 7,99 |
| $NO_3^-$ (Nitrate) mg/L | 0,74 | 1,19 | 0.52 | 0,50 | 0,91 | 0,00 | 0,96 | 0,96 | 0,67 | 0,74 |
| $NO_2^-$ (Nitrite) mg/L | 0,005 | 0,003 | 0,006 | 0,003 | 0,001 | 0,002 | 0,003 | 0,004 | 0,006 | 0,000 |
| $NH_3$ (Ammonia) mg/L | 0,03 | 0,01 | 0,06 | 0,00 | 0,03 | 0,00 | 0,03 | 0,03 | 0,06 | 0,03 |
| $PO_4^{-3}$ (Phosphate) | 0,18 | 0,09 | 0,36 | 0,08 | 0,11 | 0,18 | 0,51 | 0,21 | 0,16 | 0,21 |
| $SiO_2$ (Silica) mg/L | 12 | 5 | 6 | 7 | 6 | 6 | 8 | 11 | 6 | 12 |
| Turbidity (FAU) | 7 | 13 | 9 | 4 | 2 | 8 | 5 | 3 | 7 | 0 |

# Water Quality Sampling Data Sheet:

Date: 2/2/2015
Fishpond: Heeia

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 8:30a | % Cloud Cover | 90% |
| **Finish** Time | 2:20pm 2/2 | | |
| Tide | Medium & trapping | Precipitation | None over weekend |
| Air Temp | | Other Comments. | |
| Wind Speed/Direction | SW 10 | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 49 | 61 | 57 | 51 | 79 | 51 | 64 | 50 | 88 | 58 |
| Surf. Water Temp °C | 24.00 | 24.23 | 23.78 | 22.96 | 24.27 | 24.00 | 22.90 | 23.85 | 23.44 | 23.66 |
| Surf. Salinity | 34.50 | 34.92 | 30.35 | 25.87 | 33.71 | 34.86 | 27.61 | 34.24 | 34.28 | 34.73 |
| Surf. DO % | 59.7 | 60.0 | 58.9 | 70.1 | 59.6 | 65.2 | 75.9 | 71.6 | 79.4 | 66.6 |
| Surf. DO mg/L | 4.13 | 4.08 | 4.20 | 5.18 | 4.12 | 4.50 | 5.51 | 4.98 | 5.51 | 4.63 |
| Surf. pH | 7.76 | 7.82 | 7.71 | 7.64 | 7.78 | 7.85 | 7.73 | 7.86 | 7.86 | 7.84 |
| Bottom Temp °C | 24.07 | 24.26 | 24.31 | 24.53 | 24.32 | 24.01 | 23.90 | 23.85 | 23.47 | 23.68 |
| Bottom Salinity | 34.93 | 34.92 | 32.31 | 33.23 | 35.05 | 34.86 | 31.68 | 34.25 | 34.29 | 34.73 |
| Bottom DO% | 54.0 | 59.7 | 49.4 | 48.6 | 54.4 | 64.2 | 70.3 | 70.5 | 76.7 | 64.1 |
| Bottom DO mg/L | 3.68 | 4.10 | 3.42 | 3.31 | 3.88 | 4.43 | 4.88 | 4.83 | 5.35 | 4.51 |
| Bottom pH | 7.80 | 7.82 | 7.66 | 7.71 | 7.84 | 7.85 | 7.78 | 7.86 | 7.87 | 7.85 |
| $NO_3^-$ (Nitrate) mg/L | 0.84 | 0.96 | 0.62 | 0.51 | 1.02 | 1.40 | 0.65 | 0.88 | 0.69 | 0.42 |
| $NO_2^-$ (Nitrite) mg/L | 0.006 | 0.005 | 0.003 | 0.000 | 0.004 | 0.000 | 0.000 | 0.003 | 0.002 | 0.003 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.02 | 0.04 | 0.02 | 0.00 | 0.00 | 0.02 | 0.00 | 0.02 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.08 | 0.11 | 0.81 | 0.18 | 0.21 | 0.04 | 1.19 | 1.20 | 0.05 | 0.75 |
| $SiO_2$ (Silica) mg/L | 2 | 2 | 5 | 6 | 3 | 2 | 7 | 2 | 3 | 4 |
| Turbidity (FAU) | 2 | 0 | 2 | 4 | 0 | 0 | 3 | 3 | 0 | 4 |

# Water Quality Sampling Data Sheet:

Date: 2/5/15
Fishpond: He'eia
Initial Observations: Maury

| | | | |
|---|---|---|---|
| **Start** Time | 10:28am | % Cloud Cover | 35% |
| **Finish** Time | ~~Rising~~ 2:51pm 2/6 | | |
| Tide | low on the rise | Precipitation | 0 |
| Air Temp | | Other Comments: | Wind picked up to 15mph 7:15pm ish, 0:30 started raining light drizzle for 10min then rain falling again |
| Wind Speed/Direction | onshore in a southerly direction steady 5mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 42 | 53 | 44 | 42 | 69 | 48 | 40 | 46 | 95 | 53 |
| Surf. Water Temp °C | 22.54 | 23 83 | 25,56 | 25.60 | 24.86 | 24,80 | 25,82 | 25.17 | 25.32 | 24,05 |
| Surf. Salinity | 29,89 | 30,57 | 34.08 | 33,58 | 32,72 | 34.07 | 34,69 | [illegible] | 34.25 | 31,25 |
| Surf. DO % | 86.1 | 74.0 | 56,2 | 73,4 | 66,5 | 92.0 | ~~82.4~~ 82.7 | [illegible] | 95,5 | 93.7 |
| Surf. DO mg/L | 6.15 | 5.20 | 3,79 | 4,93 | 4,57 | 5,58 | 5,51 | [illegible] | 6,51 | 6,53 |
| Surf. pH | 7.66 | 7.82 | 7,80 | 7.81 | 7.84 | 7,86 | 7,90 | 7.90 | 7,95 | 7.89 |
| Bottom Temp °C | 22.85 | 24.65 | 25.70 | 25,75 | 25.08 | 2517 | 25.45 | 25,21 | 24,88 | 24,92 |
| Bottom Salinity | 32,05 | 33,87 | 34.92 | 34,88 | 35.09 | 34,60 | 34,67 | 34.21 | 34.26 | 32,12 |
| Bottom DO% | 80.8 | 73,8 | 63,5 | 73.7 | 68.8 | 83.5 | 85,2 | 86,7 | 95,5 | 94.5 |
| Bottom DO mg/L | 6.13 | 5,05 | 4,25 | 4,89 | 4,54 | 5.65 | 5,71 | 5.99 | 6.50 | 6,50 |
| Bottom pH | 7.85 | 7,84 | 7.82 | 7.85 | 7.88 | 7.92 | 7,89 | 7,93 | 7.95 | 7.94 |
| $NO_3^-$ (Nitrate) mg/L | 1,31 | 1,41 | 1,05 | 1,39 | 1,51 | 1.33 | 1,32 | 1,41 | 1,89 | 0.94 |
| $NO_2^-$ (Nitrite) mg/L | 0.008 | 0.004 | 0,005 | 0,003 | 0,009 | 0,016 | 0,009 | 0.004 | 0,004 | 0,005 |
| $NH_4$ (Ammonia) mg/L | 0,00 | 0,00 | 0,02 | 0,00 | 0,00 | 0.01 | 0.00 | 0,02 | 0,00 | 0,00 |
| $PO_4^{-3}$ (Phosphate) | 0,13 | 0,10 | 0,08 | 0,19 | 0,17 | 0,04 | 0,23 | 0,07 | 0,21 | 0.13 |
| $SiO_2$ (Silica) mg/L | 4 | 4 | 7 | 2 | 2 | 3 | 2 | 1 | 2 | 3 |
| Turbidity (FAU) | 5 | 3 | 10 | 5 | 0 | 2 | 12 | 7 | 1 | 4 |

# Water Quality Sampling Data Sheet:

Date: 2/9/15
Fishpond: Heeia

Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 8:10 | % Cloud Cover | 10% |
| **Finish** Time | | | |
| Tide | dropping | Precipitation | 0 |
| Air Temp | | Other Comments: | Extremely brown water not reaching from rainfall. Sites 7+8 were brown. Trailing. |
| Wind Speed/Direction | Extremely windy Constant offshore 30-40 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | | | | | | | | | | |
| Surf. Water Temp °C | 23.57 | 23.40 | 22,57 | 22,19 | 21,92 | 22.67 | 22,43 | 22,74 | 22,81 | 23,33 |
| Surf. Salinity | 32.5 | 32.41 | 28.69 | 31,87 | 29,58 | 32,61 | 34,46 | 33,68 | 33,36 | 34,33 |
| Surf. DO % | 65.1 | 69.6 | 72.6 | 85,3 | 86.3 | 82.6 | 84.8 | 82.6 | 87.7 | 82.0 |
| Surf. DO mg/L | 4.53 | 4.92 | 5.38 | 6.16 | 6.36 | 5.91 | 6.03 | 5.86 | 6.20 | 5.67 |
| Surf. pH | 7.81 | 7.83 | 7.74 | 7.88 | 7.87 | 7.88 | 7.88 | 7.87 | 7.9 | 7.89 |
| Bottom Temp °C | 23.58 | 23.40 | 22.80 | 22.38 | 22,60 | 22.67 | 22.43 | 22.75 | 22.87 | 23.34 |
| Bottom Salinity | 34.54 | 32.47 | 30.85 | 32,60 | 31,58 | 32.58 | 30.41 | 33.77 | 33.59 | 33.84 |
| Bottom DO% | 63.0 | 71.1 | 73.3 | 84.1 | 80.5 | 82.6 | 83.2 | 81.3 | | 80.4 |
| Bottom DO mg/L | 4.36 | 4.95 | 5.29 | 6.03 | 5.86 | 5.90 | 5.90 | 5.71 | 5.94 | 5.60 |
| Bottom pH | 7.82 | 7.83 | 7.82 | 7.88 | 7.86 | 7.88 | 7.87 | 7.87 | | 7.90 |
| $NO_3^-$ (Nitrate) mg/L | 1.86 | 2.24 | 0.33 | 1.02 | 0.44 | 1.39 | 1.74 | 1.56 | 1.75 | 1.55 |
| $NO_2^-$ (Nitrite) mg/L | 0.009 | 0.008 | 0.004 | 0.003 | 0.005 | 0.008 | 0.007 | 0.006 | 0.007 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.03 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.08 | 0.23 | 0.00 | 0.08 | 0.06 | 0.10 | 0.12 | 0.04 | 0.04 | 0.10 |
| $SiO_2$ (Silica) mg/L | 3 | 3 | 13 | 2 | 4 | 3 | 5 | 5 | 5 | 4 |
| Turbidity (FAU) | 8 | 7 | 13 | 12 | 2 | 8 | 148 | 140 | 15 | 2 |

10: 23.33, 34.33, 82.0, 5.67, 7.89

23.34, 34.84, 80.4, 5.60, 7.90

# Water Quality Sampling Data Sheet:

Date: 2/13/15
Fishpond: Heeia
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 9:28 | % Cloud Cover | 60 ~~60~~% but sunny |
| **Finish** Time | | | |
| Tide | | Precipitation | 0 |
| Air Temp | 76° F | Other Comments: | water becomes muddy towards sites 3-10 all makai side |
| Wind Speed/Direction | Offshore but very 20mph - 30mph gusty in various directions, ~~varies in a northerly direction~~ | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 51 | 62 | 49 | 55 | 73 | 56 | ~~32~~ 48 | 50 | 74 | 57 |
| Surf. Water Temp °C | 23.12 | 22.60 | 22,22 | 22,30 | 22.35 | 22.81 | 22.79 | 23,09 | 23,22 | 23,28 |
| Surf. Salinity | 34,50 | 30,23 | 30,40 | 30,78 | 29,85 | 34,06 | 33,29 | 32.88 | 33,93 | 30,57 |
| Surf. DO % | 72,7 | 76.5 | 85.4 | 91,5 | ~~87.6~~ 90.8 | 81.6 | 91,8 | 96,0 | 97,4 | 92,7 |
| Surf. DO mg/L | 5,11 | 5.38 | 6,25 | 6,66 | 6.71 | 5,90 | 6.52 | 6,79 | 6.85 | 6,48 |
| Surf. pH | 7,79 | 7,76 | 7,74 | 7,78 | 7,78 | 7,77 | 7,81 | 7,84 | 7,85 | 7,86 |
| Bottom Temp °C | 23,06 | 22,64 | 22,30 | 22,41 | 22.46 | 22.83 | 22.73 | 23,10 | 23,20 | 23,23 |
| Bottom Salinity | 34,50 | 30,17 | 30,33 | 32,32 | 29,90 | 31,07 | 33,30 | 32,86 | 33,94 | 34,52 |
| Bottom DO% | 73,2 | 76.0 | 87.8 | 93,9 | 89,6 | 81,7 | 91.4 | 94,1 | 97.4 | 90,1 |
| Bottom DO mg/L | 5,14 | 5,44 | 6,41 | 6,76 | 6,54 | 5,86 | 6.50 | 6,71 | 6,86 | 6,30 |
| Bottom pH | 7,80 | 7,76 | 7,77 | 7,81 | 7,78 | 7,77 | 7,81 | 7,83 | 7,87 | 7,86 |
| $NO_3^-$ (Nitrate) mg/L | 1,06 | 0,96 | 0,72 | 1,26 | 0,95 | 1,41 | 1,15 | 0,62 | 0,69 | 0,88 |
| $NO_2^-$ (Nitrite) mg/L | 0,002 | 0,003 | 0,005 | 0,000 | 0,005 | 0,004 | 0,000 | 0,002 | 0,008 | 0,002 |
| $NH_4$ (Ammonia) mg/L | 0,00 | 0,00 | 0,01 | 0,00 | 0,00 | 0,00 | 0,00 | 0,00 | 0,00 | 0,00 |
| $PO_4^{-3}$ (Phosphate) | 0,08 | 0,08 | 0,05 | 0,03 | 0,06 | 0,05 | 0,39 | 0,02 | 0,17 | 0,04 |
| $SiO_2$ (Silica) mg/L | 2 | 2 | 7 | 2 | 5 | 4 | 3 | 4 | 2 | 2 |
| Turbidity (FAU) | 0 | 5 | 10 | 10 | 3 | 3 | 29 | 28 | 11 | 6 |

# Water Quality Sampling Data Sheet:

Date: 2/24/15

Fishpond: Hi'ilo

Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 11:00am | % Cloud Cover | 30% |
| **Finish** Time | | | |
| Tide | low & falling | Precipitation | |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | Onshore 10-15 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 36 | 54 | 42 | 46 | 72 | 46 | 40 | 41 | 47 | 50 |
| Surf. Water Temp °C | 24.31 | 24.16 | 24.23 | 25.40 | 25.09 | 25.20 | 26.46 | 27.47 | 26.93 | 26.45 |
| Surf. Salinity | 18.32 | 17.92 | 17.64 | 16.93 | 15.93 | 21.17 | 22.91 | 31.01 | 30.84 | 26.05 |
| Surf. DO % | 97.1 | 95.5 | 68.6 | 94.3 | 97.2 | 96.5 | 95.2 | 96.9 | 104.7 | 97.6 |
| Surf. DO mg/L | 7.32 | 7.59 | 5.52 | 7.03 | 7.32 | 7.19 | 6.71 | 6.46 | 7.07 | 6.78 |
| Surf. pH | 7.82 | 7.81 | 7.48 | 7.74 | 7.85 | 7.91 | 7.89 | 7.97 | 8.02 | 7.99 |
| Bottom Temp °C | 27.76 | 27.75 | 28.19 | 28.05 | 28.08 | 27.93 | 28.64 | 27.51 | 27.10 | 27.53 |
| Bottom Salinity | 29.29 | 28.97 | 25.27 | 27.52 | 32.62 | 31.84 | 31.80 | 31.16 | 32.17 | 28.74 |
| Bottom DO% | 52.9 | 105.3 | 72.0 | 93.3 | 59.0 | 81.6 | 96.8 | 91.1 | 109.5 | 97.2 |
| Bottom DO mg/L | 3.55 | 7.07 | 4.88 | 6.68 | 3.85 | 5.36 | 6.31 | 6.57 | 7.35 | 6.53 |
| Bottom pH | 7.63 | 7.75 | 7.49 | 7.86 | 7.75 | 7.87 | 7.87 | 7.91 | 8.03 | 8.02 |
| $NO_3^-$ (Nitrate) mg/L | 0.86 | 0.58 | 0.33 | 0.93 | 0.71 | 0.91 | 1.37 | 1.07 | 0.72 | 1.09 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.003 | 0.000 | 0.004 | 0.000 | 0.000 | 0.002 | 0.005 | 0.000 | 0.004 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.03 | 0.02 | 0.02 | 0.00 | 0.01 | 0.04 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.16 | 0.35 | 0.22 | 0.15 | 0.19 | 0.13 | 0.14 | 0.08 | 0.06 | 0.03 |
| $SiO_2$ (Silica) mg/L | 11 | 11 | 19 | 11 | 11 | 6 | 7 | 3 | 5 | 6 |
| Turbidity (FAU) | 3 | 3 | 23 | 6 | 6 | 6 | 9 | 8 | 6 | 10 |

# Water Quality Sampling Data Sheet:

Date: 2/28/15
Fishpond: Upper

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 9:00am | % Cloud Cover | 100% |
| **Finish** Time | 10:30am | | |
| Tide | low ↓ | Precipitation | raining all day, but not heavy |
| Air Temp | cold | Other Comments: | visibility wasn't great but also not chocolate. |
| Wind Speed/Direction | 10-20 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | | | | | | | | | | |
| Surf. Water Temp °C | 23.56 | 24.27 | 25.45 | 25.44 | 25.04 | 24.12 | 24.58 | 24.05 | 24.06 | 23.86 |
| Surf. Salinity | 29.62 | 31.71 | 34.47 | 34.63 | 34.12 | 30.33 | 32.72 | 30.07 | 31.70 | 30.92 |
| Surf. DO % | 91.2 | 73.7 | 66.4 | 69.1 | 80.3 | 85.1 | 76.2 | 81.0 | 83.7 | 142.9 |
| Surf. DO mg/L | 6.53 | 5.15 | 3.13 | 4.67 | 5.46 | 6.01 | 5.28 | 5.60 | 5.74 | 10.11 |
| Surf. pH | 7.89 | 7.82 | 7.71 | 7.72 | 7.88 | 7.88 | 7.81 | 7.83 | 7.84 | 7.81 |
| Bottom Temp °C | 24.02 | 25.27 | 25.18 | 25.48 | 25.15 | 24.91 | 25.54 | 24.23 | 24.30 | 23.99 |
| Bottom Salinity | 29.96 | 34.57 | 34.76 | 34.55 | 34.79 | 33.09 | 33.92 | 30.09 | 32.66 | 31.36 |
| Bottom DO% | 62.8 | 44.4 | 37.6 | 53.5 | 57.3 | 76.7 | 55.4 | 60.8 | 82.4 | 92.0 |
| Bottom DO mg/L | 5.67 | 3.13 | 3.21 | 3.60 | 3.87 | 5.26 | 3.79 | 5.70 | 6.06 | 6.47 |
| Bottom pH | 7.82 | 7.74 | 7.72 | 7.75 | 7.81 | 7.83 | 7.78 | 7.83 | 7.90 | 7.88 |
| $NO_3^-$ (Nitrate) mg/L | 1.03 | 1.09 | 1.02 | 1.31 | 1.11 | 1.14 | 1.03 | 1.12 | 1.12 | 1.12 |
| $NO_2^-$ (Nitrite) mg/L | 0.001 | 0.000 | 0.002 | 0.002 | 0.000 | 0.004 | 0.001 | 0.000 | 0.000 | 0.001 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.08 | 0.07 | 0.06 | 0.07 | 0.14 | 0.05 | 0.00 | 0.12 | 0.07 | 0.08 |
| $SiO_2$ (Silica) mg/L | 5 | 4 | 6 | 3 | 3 | 4 | 5 | 5 | 5 | 4 |
| Turbidity (FAU) | 3 | 12 | 14 | 9 | 3 | 4 | 3 | 6 | 4 | 4 |

# Water Quality Sampling Data Sheet:

Date: 3/3/15
Fishpond: Hefia

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 10:30 am | % Cloud Cover | 80% |
| **Finish** Time | 11:40 am | | |
| Tide | high ↑ | Precipitation | light drizzle halfway through, then rained lightly the rest |
| Air Temp | | Other Comments: | for the amount of rain we had before & today the water was not chocolately. I was surprised. |
| Wind Speed/Direction | calm consistent | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 43 | 54 | 46 | 50 | 75 | 49 | 45 | 48 | 56 | 58 |
| Surf. Water Temp °C | 23.93 | 24.00 | 23.39 | 20.70 | 25.35 | 25.12 | 25.40 | 25.17 | 24.83 | 25.42 |
| Surf. Salinity | 21.76 | 23.09 | 8.36 | 21.21 | 25.50 | 32.35 | 33.3 | 30.52 | 33.7 | 32.11 |
| Surf. DO % | 107.0 | 96.3 | 85.5 | 91.4 | 92.5 | 100.2 | 93.9 | 101.2 | 94.2 | 97.5 |
| Surf. DO mg/L | 7.93 | 7.09 | 6.74 | 6.50 | 6.66 | 6.88 | 6.75 | 7.02 | 6.75 | 6.42 |
| Surf. pH | 7.79 | 7.79 | 7.67 | 7.68 | 7.81 | 7.88 | 7.87 | 7.84 | 7.87 | 7.58 |
| Bottom Temp °C | 23.98 | 24.83 | 25.24 | 25.54 | 25.54 | 25.15 | 25.62 | 25.19 | 24.88 | 25.60 |
| Bottom Salinity | 21.82 | 24.16 | 28.26 | 30.89 | 30.40 | 32.35 | 33.70 | 30.49 | 33.40 | 32.03 |
| Bottom DO% | 104.0 | 96.4 | 59.1 | 92.2 | 6.06 | 100.7 | 95.8 | 101.5 | 98.7 | 93.6 |
| Bottom DO mg/L | 7.73 | 6.77 | 4.13 | 6.88 | 6.06 | 6.90 | 6.58 | 7.03 | 6.76 | 6.39 |
| Bottom pH | 7.82 | 7.83 | 7.86 | 7.74 | 7.77 | 7.89 | 7.89 | 7.88 | 7.88 | 7.55 |
| $NO_3^-$ (Nitrate) mg/L | 1.11 | 0.85 | 0.36 | 1.16 | 0.69 | 0.98 | 0.95 | 1.02 | 1.34 | 0.83 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.002 | 0.000 | 0.001 | 0.001 | 0.000 | 0.004 | 0.005 | 0.004 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.03 | 0.01 | 0.00 | 0.00 | 0.02 | 0.00 | 0.00 | 0.02 |
| $PO_4^{-3}$ (Phosphate) | 0.09 | 0.00 | 0.08 | 0.00 | 0.76 | 0.15 | 0.27 | 0.07 | 0.35 | 0.06 |
| $SiO_2$ (Silica) mg/L | 6 | 7 | 21 | 8 | 7 | 3 | 5 | 4 | 2 | 3 |
| Turbidity (FAU) | 3 | 2 | 24 | 18 | 2 | 0 | 6 | 3 | 0 | 7 |

blue??

# Water Quality Sampling Data Sheet

Date: 3/12/5

Fishpond: He°ei'a

Initial Observations Maury Lake, Justin

| | | | |
|---|---|---|---|
| Start Time | 9:20 pm | % Cloud Cover | 70% |
| Finish Time | | | |
| Tide | high ↓ | Precipitation | light drizzle forecast |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | Onshore northerly 5-10mph | | |

| Site# | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 76 | 83 | 71 | 73 | 93 | 73 | 70 | 72 | 82 | 68 |
| Surf Water Temp | 22,11 | 22,30 | 23.50 | 23.64 | 21,90 | 22.39 | 23,47 | 22,64 | 22,22 | 21,36 |
| Surf Salinity | 32,54 | 32.30 | 31.05 | 32.00 | 30,05 | 32,13 | 31,90 | 32,23 | 31.58 | 33,23 |
| Surf D.O.% | 104.0 | 94.3 | 84.4 | 98.1 | 95,9 | 94,1 | 93,7 | 99.5 | 93.9 | 98.5 |
| Surf D.O. mg/L | 7.43 | 6.82 | 6,00 | 6.91 | 7,08 | 6.77 | 6.62 | 7,13 | 6.79 | 7,16 |
| Surf pH | 7,98 | 8,00 | 7,91 | 8.02 | 8.02 | 8.02 | 7.91 | 8.03 | 8.02 | 8.04 |
| Bottom Temp (C) | 2237 | 22.33 | 23,70 | 23.58 | 22.50 | 22.38 | 23.80 | 22,74 | 22 25 | 21.27 |
| Bottom Salinity | 33.97 | 33.74 | 32.25 | 32.58 | 33,35 | 33.74 | 33,31 | 32,39 | 32.11 | 33,69 |
| Bottom D.O. % | 101.8 | 94.2 | 86.6 | 100.3 | 94.87 | 91.0 | 84.0 | 94.2 | 86.4 | 96.4 |
| Bottom D.O. mg/L | 7.24 | 6.71 | 6,34 | 7.05 | 6.74 | 6.47 | 6.15 | 6,72 | 6.16 | 7,03 |
| Bottom pH | 8,00 | 7.96 | 8.04 | 8.04 | 8,00 | 8.00 | 7.98 | 8.02 | 7.98 | 8,03 |
| NO3 (Nitrate) mg/L | 1.66 | 1.17 | 1,48 | 0,37 | 0,08 | 1,10 | 1.39 | 1,67 | 0.98 | 1,21 |
| NO2 (Nitrite) mg/L | 0,006 | 0,005 | 0,010 | 0.002 | 0.002 | 0.004 | 0.004 | 0,005 | 0,000 | 0,003 |
| NH4 (Ammonia) mg/L | 0,00 | ~~0.00~~ | 0,00 | 0,00 | ---- | 0,00 | 0,00 | 0,00 | 0,00 | ---- |
| PO4 (Phosphate) | 0.14 | 0,11 | 0,10 | 0.09 | 0.05 | 0.03 | 0,08 | 0,15 | 0,20 | 0.17 |
| SiO2 (Silica) mg/L | 2 | 3 | 4 | 7 | 4 | 7 | 2 | 2 | 3 | 2 |
| Turbidity (FAU) | 12 | 5 | 8 | 14 | 3 | 4 | 8 | 2 | 1 | 4 |

# Water Quality Sampling Data Sheet

Date: 3/15/15

Fishpond: Heʻeia

Initial Observations Hourly

| | | | |
|---|---|---|---|
| Start Time | 2:22 pm | % Cloud Cover | 50% clear |
| Finish Time | | | |
| Tide | low ↓ | Precipitation | |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | light winds onshore 5 mph | | |

| Site# | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 40 | 52 | 43 | 48 | 62 | 44 | 40 | 41 | 59 | 42 |
| Surf Water Temp | 21.39 | 21.73 | 21.16 | 27.19 | 23.06 | 23.00 | 23.15 | 25.33 | 23.05 | 22.11 |
| Surf Salinity | 17.24 | 24.78 | 24.91 | 26.10 | 30.50 | 32.40 | 30.76 | 30.63 | 33.06 | 27.10 |
| Surf D.O.% | 104.5 | 104.2 | 91.8 | 105.4 | 118.7 | 127.6 | 136.9 | 118.9 | 129.7 | 106.9 |
| Surf D.O. mg/L | 6.33 | 7.75 | 6.09 | 7.79 | 8.54 | 9.08 | 9.62 | 8.48 | 9.23 | 8.20 |
| Surf pH | 7.71 | 7.96 | 7.80 | 7.93 | 8.05 | 8.08 | 8.02 | 8.06 | 8.13 | 7.92 |
| Bottom Temp (C) | 22.47 | 22.68 | 22.96 | 23.17 | 23.52 | 23.00 | 23.25 | 23.39 | 23.18 | 23.62 |
| Bottom Salinity | 20.21 | 30.78 | 30.52 | 30.97 | 31.56 | 32.59 | 30.65 | 30.66 | 33.08 | 32.85 |
| Bottom D.O. % | 116.0 | 114.7 | 100.5 | 109.6 | 118.3 | 120.2 | 122.6 | 118.7 | 133.7 | 116.3 |
| Bottom D.O. mg/L | 6.84 | 8.30 | 7.53 | 7.85 | 8.38 | 9.19 | 8.74 | 8.48 | 9.46 | 8.35 |
| Bottom pH | 7.91 | 8.02 | 7.92 | 7.98 | 8.04 | 8.09 | 8.04 | 8.07 | 8.13 | 8.04 |
| NO3 (Nitrate) mg/L | | | | | | | | | | |
| NO2 (Nitrite) mg/L | | | | | | | | | | |
| NH4 (Ammonia) mg/L | | | This data on Next Sheet! | | | | | | | |
| PO4 (Phosphate) | | | | | | | | | | |
| SiO2 (Silica) mg/L | | | | | | | | | | |
| Turbidity (FAU) | | | | | | | | | | |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Bottom pH | | 15 | | | | | | | | |
| NO3 (Nitrate) mg/L | 1.25 | 1.46 | 0.00 | 0.42 | 0.22 | 1.28 | 0.96 | 1.20 | 1.56 | 1.36 |
| NO2 (Nitrite) mg/L | 0.010 | 0.006 | 0.000 | 0.003 | 0.000 | 0.003 | 0.010 | 0.000 | 0.003 | 0.003 |
| NH4 (Ammonia) mg/L | 0.00 | 0.00 | 0.00 | 0.00 | 0.04 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| PO4 (Phosphate) | 0.32 | 0.19 | 0.11 | 0.08 | 0.11 | 0.30 | 0.08 | 0.12 | 0.10 | 0.07 |
| SiO2 (Silica) mg/L | 2 | 2 | 12 | 7 | 12 | 2 | 3 | 1 | 2 | 2 |
| Turbidity (FAU) | 4 | 0 | 8 | 3 | 3 | 2 | 0 | 2 | 0 | 3 |

Date: 3-16-15

Fishpond: Heeia

Initial Observations

| | | | |
|---|---|---|---|
| Start Time | 9:00-10:30 | % Cloud Cover | 40% Sunny bright |
| Finish Time | | | |
| Tide | ~~Low~~ & Rising | Precipitation | None |
| Air Temp | 80° | Other Comments: | |
| Wind Speed/Direction | ~~0-5 None~~ NE | | |

| Site# | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 16 | 20 | 22 | ~~17~~ | 26 | 19 | 16 | 18 | 24 | 22.5 |
| Surf Water Temp | ~~no~~ | | | | | | | | | |
| Surf Salinity | ~~o~~ | all | | around | | 30 | ppt | | | |
| Surf D.O.% | | | | collected | | | | | | |
| Surf D.O. mg/L | | | | this data | | | | | | |
| Surf pH | | | | w/ EXO | | | | | | |
| Bottom Temp (C) | | | | | | | | | | |
| Bottom Salinity | | | | | | | | | | |
| Bottom D.O. % | | | | | | | | | | |
| Bottom D.O. mg/L | ↓ | | | Samples in Fridge. | | | | | ↓ | |
| Bottom pH | | | | | | | | | | |
| NO3 (Nitrate) mg/L | 0.41 | 0.18 | 0.00 | 0.51 | 0.13 | 1.03 | 1.19 | 0.00 | 1.01 | 0.29 |
| NO2 (Nitrite) mg/L | 0.000 | 0.006 | 0.000 | 0.004 | 0.000 | 0.000 | 0.002 | 0.009 | 0.000 | 0.000 |
| NH4 (Ammonia) mg/L | 0 | 8 | 23 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| PO4 (Phosphate) | 0.14 | 0.16 | 0.12 | 0.15 | 0.02 | 0.19 | 0.47 | 0.10 | 0.14 | 0.09 |
| SiO2 (Silica) mg/L | 8 | 10 | 18 | 5 | 3 | 0 | 3 | 3 | 2 | 7 |
| Turbidity (FAU) | 2 | 5 | 11 | 5 | 7 | 1 | 4 | 0 | 0 | 0 |

DO analysis

# Water Quality Sampling Data Sheet

Date: 3/22/15

Fishpond: He'eia

Initial Observations Mauka

| | | | |
|---|---|---|---|
| Start Time | 5:23pm | % Cloud Cover | 60% |
| Finish Time | | | |
| Tide | High ↓ | Precipitation | |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | onshore 10 mph (constant) | | |

| Site# | 1 | 2 | 6 | 4 | 5 | 3 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 80 | 92 | 86 | 77 | ~110 | 82 | 80 | 82 | 81 | 91 |
| Surf Water Temp | 26.71 | 27.32 | 26.75 | 26.63 | 26.84 | 24.86 | 27.62 | 25.61 | 24.74 | 25.64 |
| Surf Salinity | 27.32 | 29.41 | 31.06 | 20.72 | 23.84 | 11.08 | 27.40 | 34.32 | 34.35 | 34.22 |
| Surf D.O.% | 114.0 | 118.2 | 116.7 | 102.7 | 110.8 | 86.9 | 110.3 | 110.6 | 105.3 | 114.9 |
| Surf D.O. mg/L | 7.83 | 7.96 | 7.85 | 7.35 | 7.76 | 6.71 | 7.49 | 7.45 | 7.21 | 7.75 |
| Surf pH | 8.05 | 8.18 | 8.21 | 8.04 | 8.13 | 7.86 | 8.16 | 8.18 | 8.14 | 8.29 |
| Bottom Temp (C) | 27.11 | 26.69 | 25.23 | 28.07 | 27.75 | 28.78 | 26.73 | 25.71 | 24.71 | 25.70 |
| Bottom Salinity | 33.77 | 33.82 | 34.14 | 28.99 | 33.34 | 30.85 | 33.19 | 34.19 | 34.33 | 34.20 |
| Bottom D.O. % | 132.2 | 116.7 | 114.5 | 112.5 | 119.66 | 105.8 | 116.6 | 112.4 | 107.6 | 119.3 |
| Bottom D.O. mg/L | 8.71 | 7.74 | 7.72 | 7.38 | 7.92 | 6.88 | 7.75 | 7.56 | 7.36 | 8.03 |
| Bottom pH | 8.19 | 8.16 | 8.13 | 8.12 | 8.19 | 8.07 | 8.15 | 8.18 | 8.14 | 8.25 |
| NO3 (Nitrate) mg/L | 0.28 | 0.03 | 0.05 | 0.27 | 0.92 | 0.00 | 0.96 | 0.87 | 0.79 | 1.92 |
| NO2 (Nitrite) mg/L | 0.000 | 0.000 | 0.000 | 0.003 | 0.006 | 0.000 | 0.002 | 0.003 | 0.002 | 0.002 |
| NH4 (Ammonia) mg/L | 0.00 | 0.00 | 0.02 | 0.00 | 0.20 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| PO4 (Phosphate) | 0.00 | 0.07 | 0.05 | 0.00 | 0.17 | 0.00 | 0.19 | 0.04 | 0.05 | 0.02 |
| SiO2 (Silica) mg/L | 3 | 3 | 3 | 6 | 7 | 12 | 5 | 0 | 1 | 1 |
| Turbidity (FAU) | 5 | 7 | 19 | 2 | 4 | 3 | 5 | 9 | 8 | 2 |

# Water Quality Sampling Data Sheet:

Date: 4/11/15
Fishpond: Heleia
Initial Observations: Hazy

| | | | |
|---|---|---|---|
| **Start** Time | 1042 | % Cloud Cover | 80 |
| **Finish** Time | 2330 | | |
| Tide | Low | Precipitation | 0 |
| Air Temp | 23.30 | Other Comments: | |
| Wind Speed/Direction | onshore 15 MPH | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 2ft | 1.5ft | 1.5ft | 1.0ft | 3ft | 1.5ft | 1.8ft | 2ft | 2ft | 2.3ft |
| Surf. Water Temp °C | 24.98 | 25 | 24.98 | 25.14 | 24.75 | 25.8 | 26.28 | 25.77 | 26.62 | 26.74 |
| Surf. Salinity | 22.61 | 20.44 | 11.77 | 16.82 | 23.4 ~~33~~ | 28.88 | 24.54 | 31.23 | 27.54 | 26.59 |
| Surf. DO % | 95.0 | 96.6 | ~~78.1~~ | 91.1 | 92.9 | 98.4 | 98.5 | 124.9 | 112.6 | 109.1 |
| Surf. DO mg/L | 6.85 | 7.10 | 5.96 | 6.81 | 7.02 | 6.83 | 6.93 | 8.30 | 7.6 | 7.51 |
| Surf. pH | 7.80 | 7.99 | 7.73 | 7.88 | 8.00 | 8.08 | 8.04 | 8.11 | 8.19 | 8.13 |
| Bottom Temp °C | ~~25.47~~ | ~~27.50~~ | 27.07 | ~~26.33~~ | 26.36 | 25.92 | 26.28 | 25.82 | 26.62 | 26.73 |
| Bottom Salinity | 24.10 | 26.99 | 22.42 | 24.21 | 31.53 | 28.88 | 24.37 | 31.29 | 27.85 | 26.56 |
| Bottom DO% | 91.5 | 100.5 | 82.8 | 101.4 | ~~67.9~~ | 100.0 | 100.5 | 117.0 | 116.0 | 109.0 |
| Bottom DO mg/L | 6.30 | 6.78 | 5.80 | 7.18 | 4.57 | 7.04 | 7.05 | 7.91 | 7.98 | 7.53 |
| Bottom pH | 7.90 | 8.07 | 7.91 | 8.03 | 7.93 | 8.09 | 8.04 | 8.11 | 8.19 | 8.13 |
| $NO_3^-$ (Nitrate) mg/L | 1.02 | 1.14 | 1.25 | 0.77 | 0.83 | 0.00 | 1.15 | 0.72 | 0.58 | 0.70 |
| $NO_2^-$ (Nitrite) mg/L | 0.003 | 0.000 | 0.000 | 0.000 | 0.000 | 0.003 | 0.000 | 0.000 | 0.002 | 0.008 |
| $NH_4$ (Ammonia) mg/L | 0.02 | 0.03 | 0.03 | 0.02 | 0.05 | 0.02 | 0.03 | 0.00 | 0.02 | 0.02 |
| $PO_4^{-3}$ (Phosphate) | 0.27 | 0.04 | 0.07 | 0.10 | 0.02 | 0.17 | 0.29 | 0.17 | 0.11 | 0.08 |
| $SiO_2$ (Silica) mg/L | 4/15 | Having | realised | I ran | the | Ammonia | test | in | steq. | Sorry |
| Turbidity (FAU) | 5 | 4 | 21 | 8 | 5 | 7 | 17 | 2 | 4 | 4 |

# Water Quality Sampling Data Sheet:

Date: 4/13/15
Fishpond: Heeia Maury
Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 9:38 am | % Cloud Cover | 40% mostly mauka |
| **Finish** Time | ~~4:53~~ | | |
| Tide | low ↑ | Precipitation | 0 |
| Air Temp | | Other Comments: | very sunny day light vog good visibility |
| Wind Speed/Direction | onshore 5 mph 10 mph gusts | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 1.25ft | 2 ft | 2.3ft | 1.8ft | 3.5ft | 1.75ft | 1.3ft | 1.4ft | 2.3ft | 2.25ft |
| Surf. Water Temp °C | 25.16 | 24.31 | 25.01 | 25.43 | 25.20 | 26.19 | 25.50 | 25.95 | 25.60 | 26.26 |
| Surf. Salinity | 23.68 | 17.50 | ~~15~~ 15.88 | 22.36 | 21.93 | 29.02 | 29.72 | 29.88 | 32.75 | 28.62 |
| Surf. DO % | 98.2 | 95.4 | 82.7 | 95.6 | 93.6 | 106.1 | 96.7 | 104.7 | 109.8 | 104.5 |
| Surf. DO mg/L | 7.03 | 7.21 | ~~6~~ 6.22 | 6.91 | 6.81 | 7.27 | 6.87 | ~~7.19~~ | 7.48 | 7.18 |
| Surf. pH | 7.98 | 7.92 | 7.66 | 7.94 | 7.99 | 8.08 | 7.97 | 8.06 | 8.09 | 8.09 |
| Bottom Temp °C | 25.17 | ~~24.61~~ | 25.58 | 25.51 | 26.49 | 26.19 | 25.84 | 25.95 | 25.71 | 26.28 |
| Bottom Salinity | 23.71 | ~~22.13~~ 19.50 | 19.23 | 23.98 | 29.99 | 29.72 | 23.80 | 29.92 | 32.95 | 28.71 |
| Bottom DO% | 92.0 | 93.4 | 83.9 | 94.9 | 78.8 | 105.9 | 96.1 | 104.8 | 113.5 | 105.2 |
| Bottom DO mg/L | 6.72 | 6.95 | 6.11 | 6.78 | 5.35 | 7.26 | 6.83 | 7.18 | 7.71 | 7.22 |
| Bottom pH | 7.98 | 8.01 | 7.76 | 7.98 | 7.92 | 8.07 | 7.98 | 8.06 | 8.10 | 8.09 |
| $NO_3^-$ (Nitrate) mg/L | 0.33 | 0.15 | 0.00 | 0.62 | 0.20 | 1.21 | 0.37 | 0.74 | 1.60 | 0.98 |
| $NO_2^-$ (Nitrite) mg/L | 0.002 | 0.005 | 0.009 | 0.005 | 0.004 | 0.009 | 0.000 | 0.003 | 0.000 | 0.000 |
| $NH_3$ (Ammonia) mg/L | 0.00 | 0.00 | 0.02 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 |
| $PO_4^{-3}$ (Phosphate) | 0.04 | 0.03 | 0.13 | 0.12 | 0.03 | 0.03 | 0.02 | 0.00 | 0.00 | 0.00 |
| $SiO_2$ (Silica) mg/L | 7 | 10 | 15 | 6 | 8 | 3 | 4 | 3 | 2 | 3 |
| Turbidity (FAU) | 17 | 3 | 21 | 8 | 4 | 8 | 10 | 5 | 0 | 0 |

# Water Quality Sampling Data Sheet:

Date: 4/20/15 Heeia

Fishpond:

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 8:30am | % Cloud Cover | 60 |
| **Finish** Time | 9:15am | | |
| Tide | Low | Precipitation | ∅ |
| Air Temp | 80 | Other Comments: | |
| Wind Speed/Direction | ENE | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | | | | | | | | | | |
| Surf. Water Temp °C | 22.93 | 22.68 | 22.83 | 23.03 | 23.09 | 22.65 | 23.24 | 22.79 | 22.98 | 23.23 |
| Surf. Salinity | 28.39 | 24.90 | 20.85 | 27.14 | 24.57 | 31.04 | 26.75 | 28.98 | 32.03 | 29.37 |
| Surf. DO % | 91.3 | 91.7 | 86.4 | 89.8 | 86.6 | 96.7 | 88.6 | 99.3 | 99.9 | 94.8 |
| Surf. DO mg/L | 6.63 | 6.85 | 6.61 | 6.56 | 6.46 | 6.95 | 6.45 | 7.24 | 7.13 | 6.81 |
| Surf. pH | 7.86 | 7.91 | 7.77 | 7.84 | 7.93 | 7.99 | 7.89 | 7.97 | 8.01 | 7.97 |
| Bottom Temp °C | 22.94 | 23.15 | 22.89 | 23.11 | 23.25 | 22.76 | 23.24 | 22.81 | 23.00 | 23.26 |
| Bottom Salinity | 28.73 | 27.35 | 20.68 | 27.21 | 26.15 | 31.02 | 26.95 | 28.95 | 32.14 | 29.36 |
| Bottom DO% | 89.3 | 90.0 | 86.9 | 87.2 | 77.3 | 95.7 | 86.3 | 99.3 | 101.3 | 92.6 |
| Bottom DO mg/L | 6.49 | 6.63 | 6.61 | 6.35 | 5.56 | 6.89 | 6.32 | 7.24 | 7.20 | 6.66 |
| Bottom pH | 7.89 | 7.92 | 7.73 | 7.90 | 7.87 | 8.00 | 7.90 | 7.99 | 8.02 | 7.96 |
| $NO_3^-$ (Nitrate) mg/L | 0.94 | 0.10 | 0.62 | 0.41 | 0.67 | 0.78 | 0.32 | 1.12 | 0.81 | 0.75 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.005 | 0.000 | 0.002 | 0.002 | 0.000 | 0.000 | 0.003 | 0.004 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 | 0.00 | 0.01 | 0.00 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.01 | 0.09 | 0.05 | 0.10 | 0.06 | 0.00 | 0.00 | 0.04 | 0.03 | 0.00 |
| $SiO_2$ (Silica) mg/L | 3 | 7 | 9 | 6 | 7 | 2 | 3 | 4 | 2 | 5 |
| Turbidity (FAU) | 17 | 5 | 13 | 18 | 11 | [illegible] | 23 | 9 | 0 | 9 |

3

# Water Quality Sampling Data Sheet:

Date: 4/21/15
Fishpond: Heeia

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 8:25 | % Cloud Cover | 90 %. Lots other Kaolaus relatively clear over ocean |
| **Finish** Time | 9:45 | | |
| Tide | low & dropping | Precipitation | 0% but some last night |
| Air Temp | 80° | Other Comments: | |
| Wind Speed/Direction | ENE 10-15 | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 40 | 48 | 46 | 45 | 65 | 46 | 36 | 40 | 43 | 45 |
| Surf. Water Temp °C | 22.31 | 22.22 | 22.48 | 22.50 | 22.65 | 22.30 | 22.90 | 22.55 | 22.75 | 22.68 |
| Surf. Salinity | 26.38 | 24.12 | 17.66 | 27.20 | 28.35 | 32.41 | 27.19 | 29.98 | 32.88 | 27.44 |
| Surf. DO % | 90.2 | 90.9 | 84.4 | 93.3 | 85.0 | 97.5 | 89.1 | 100.7 | 99.7 | 95.4 |
| Surf. DO mg/L | 6.74 | 6.87 | 6.59 | 6.90 | 6.22 | 7.02 | 6.55 | 7.33 | 7.11 | 7.02 |
| Surf. pH | 7.88 | 7.88 | 7.67 | 7.94 | 7.94 | 8.00 | 7.90 | 8.03 | 8.03 | 7.99 |
| Bottom Temp °C | 22.33 | 22.34 | 22.58 | 22.53 | 22.70 | 22.36 | 22.87 | 22.60 | 22.79 | 22.72 |
| Bottom Salinity | 26.42 | 24.59 | 21.38 | 27.24 | 28.73 | 32.62 | 27.17 | 30.15 | 33.03 | 27.40 |
| Bottom DO% | 86.6 | 88.4 | 82.6 | 91.2 | 84.1 | 97.4 | 86.9 | 97.3 | 98.7 | 92.9 |
| Bottom DO mg/L | 6.44 | 6.65 | 6.33 | 6.76 | 6.11 | 7.02 | 6.39 | 7.06 | 7.03 | 6.86 |
| Bottom pH | 7.89 | 7.90 | 7.75 | 7.94 | 7.96 | 8.01 | 7.89 | 8.04 | 8.04 | 7.98 |
| $NO_3^-$ (Nitrate) mg/L | 0.59 | 1.35 | 0.71 | 0.61 | 0.30 | 1.54 | 0.65 | 0.81 | 1.05 | 0.53 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.003 | 0.001 | 0.008 | 0.002 | 0.004 | 0.003 | 0.000 | 0.007 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.05 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.20 |
| $PO_4^{-3}$ (Phosphate) | 0.00 | 0.11 | 0.08 | 0.06 | 0.05 | 0.16 | 0.17 | 0.08 | 0.00 | 0.05 |
| $SiO_2$ (Silica) mg/L | 6 | 6 | 9 | 5 | 4 | 3 | 5 | 3 | 2 | 4 |
| Turbidity (FAU) | 23 | 9 | 20 | 9 | 16 | 9 | 42 | 16 | 1 | 10 |

# Water Quality Sampling Data Sheet:

Date: 4/29/15
Fishpond: He'eia
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 2:30 | % Cloud Cover | 80% but sunny |
| **Finish** Time | | | |
| Tide | High | Precipitation | 0 |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | 20-25mph onshore constant | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 54 | 65 | 65 | 54 | 76 | 57 | 46 | 47 | 53 | 58 |
| Surf. Water Temp °C | 27,17 | 26,23 | 25,47 | 25,99 | 23,81 | 26.56 | 25,95 | 25,78 | 26.06 | 26,79 |
| Surf. Salinity | 26,99 | 23,16 | 22,30 | 27,18 | 26,29 | 32.68 | 27,00 | 33,72 | 30,01 | 27,69 |
| Surf. DO % | 121,9 | 107,2 | 160,2 | 107,1 | 121,3 | 118,1 | 100,6 | 107,7 | 117,9 | 104,9 |
| Surf. DO mg/L | 8,33 | 7,61 | 10,15 | 7,53 | 8,31 | 7,88 | 7,01 | 7,09 | 8,33 | 7,23 |
| Surf. pH | 7,96 | 7,99 | 7,99 | 8,07 | 8,01 | 8,06 | 8,00 | 8,07 | 8,12 | 8,10 |
| Bottom Temp °C | 27,19 | 26,41 | 25,45 | 25,95 | 26,02 | 26,57 | 26,04 | 25,51 | 26,22 | 26,55 |
| Bottom Salinity | 26,94 | 29,40 | 21,76 | 27,23 | 30,10 | 32,06 | 26,93 | 33,78 | 30,11 | 27,44 |
| Bottom DO% | 118,3 | 111,2 | 110,3 | 108,7 | 112,6 | 115,3 | 101,6 | 106,8 | 111,4 | 106,6 |
| Bottom DO mg/L | 8,08 | 7,70 | 7,97 | 7,55 | 9,72 | 7,70 | 7,07 | 7,72 | 7,73 | 7,30 |
| Bottom pH | 7,98 | 8,00 | 7,96 | 8,09 | 8,07 | 8,07 | 8,00 | 8,09 | 8,12 | 8,10 |
| $NO_3^-$ (Nitrate) mg/L | 0,99 | 1,06 | 0.56 | 0.70 | 0,97 | 1,22 | 0,73 | 1,62 | 1,04 | 0.31 |
| $NO_2^-$ (Nitrite) mg/L | 0.002 | 0.006 | 0.000 | 0.003 | 0.006 | 0.006 | 0,000 | 0.000 | 0,000 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0,00 | 0.05 | 0,01 | 0,00 | 0.04 | 0,00 | 0.03 | 0.00 | 0.02 | 0.03 |
| $PO_4^{-3}$ (Phosphate) | 0.09 | 0.09 | 0.13 | 0.07 | 0.10 | 0.12 | 0.05 | 0.06 | 0.07 | 0,00 |
| $SiO_2$ (Silica) mg/L | 5 | 6 | 10 | 5 | 6 | 2 | 3 | 2 | 2 | 3 |
| Turbidity (FAU) | 13 | 6 | 35 | 21 | 8 | 11 | 19 | 0 | 3 | 21 |

# Water Quality Sampling Data Sheet:

Date: 5/4/15
Fishpond: Heeia
Initial Observations: Maury

| | | | |
|---|---|---|---|
| **Start** Time | 3:45 pm | % Cloud Cover | 75% Sunny |
| **Finish** Time | | | |
| Tide | High ↑ | Precipitation | 0 |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | Onshore 10 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 83 | 92 | 77 | 82 | ~~200~~ ~100 | 81 | 76 | 78 | 86 | 87 |
| Surf. Water Temp °C | 26.51 | 26.52 | 24.63 | 26.07 | 26.38 | 26.26 | 26.30 | 26.10 | 25.52 | 26.04 |
| Surf. Salinity | 22.98 | 23.52 | 12.92 | 21.99 | 20.19 | 24.96 | 26.45 | 30.05 | 35.03 | 34.79 |
| Surf. DO % | 110.8 | 117.4 | 87.1 | 107.9 | 118.2 | 112.2 | 107.6 | 113.1 | 104.1 | ~~3.6~~ 113.6 |
| Surf. DO mg/L | 7.85 | 8.18 | 6.63 | 7.74 | 8.25 | 7.88 | 7.52 | 8.02 | 6.99 | 7.58 |
| Surf. pH | 7.91 | 8.11 | 7.73 | 7.99 | 8.07 | 8.11 | 8.05 | ~~8.06~~ 8.06 | 8.04 | 8.09 |
| Bottom Temp °C | 26.66 | 26.59 | 26.72 | 26.76 | 26.57 | ~~25.97~~ 26.43 | 26.43 | ~~26.17~~ 26.17 | 25.59 | 26.12 |
| Bottom Salinity | ~~39.52~~ 34.52 | ~~34.40~~ 34.40 | 29.09 | ~~31.47~~ 31.47 | 34.36 | 34.93 | ~~32.56~~ 32.56 | 34.14 | 35.04 | 34.81 |
| Bottom DO% | 131.6 | 117.5 | 101.5 | ~~105.9~~ 105.9 | 111.8 | 110.2 | 113.9 | ~~110.8~~ 110.8 | 105.1 | 115.8 |
| Bottom DO mg/L | 8.68 | 7.76 | 6.86 | 7.17 | 7.45 | 7.33 | 7.60 | ~~7.40~~ 7.40 | 7.05 | 7.72 |
| Bottom pH | 8.06 | 8.01 | 7.95 | 7.99 | 8.03 | 7.99 | 8.02 | 8.04 | 8.01 | 8.09 |
| $NO_3^-$ (Nitrate) mg/L | 1.15 | 0.29 | 0.60 | 0.71 | 0.93 | 1.11 | 0.64 | 0.73 | 0.87 | 1.49 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.001 | 0.000 | 0.001 | 0.002 | 0.002 | 0.005 | 0.000 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.01 | 0.02 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.00 | 0.00 | 0.16 | 0.08 | 0.06 | 0.02 | 0.07 | 0.12 | 0.00 | 0.03 |
| $SiO_2$ (Silica) mg/L | 6 | 4 | 11 | 6 | 4 | 5 | 4 | 2 | 0 | 0 |
| Turbidity (FAU) | 6 | 2 | 9 | 4 | 8 | 2 | 3 | 0 | 0 | 0 |

5: 110.5

# Water Quality Sampling Data Sheet:

Date: 5/20/15
Fishpond: Heeia

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 1:51pm | % Cloud Cover | 90% |
| **Finish** Time | | | |
| Tide | High↑ | Precipitation | Showers earlier today |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | Onshore | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 56 | 69 | 68 | 73 | 78 | 64 | 69 | 72 | 79 | 85 |
| Surf. Water Temp °C | 25,62 | 25.51 | 24,12 | 25.98 | 25,43 | 24,93 | 25.71 | 24,72 | 24.93 | 24,94 |
| Surf. Salinity | 26.54 | 26.71 | 10,00 | 23.80 | 24,83 | 33,68 | 28,53 | 34,51 | 34.42 | 34,34 |
| Surf. DO % | 106.1 | 108.4 | 69,5 | 95,2 | 104.9 | 99,8 | 101.9 | 89.2 | 94,3 | 92,1 |
| Surf. DO mg/L | 7.40 | 7,63 | 5,45 | 6,86 | 7,53 | 6,82 | 7,09 | 6,08 | 6,43 | 6,26 |
| Surf. pH | 8.00 | 8,01 | 7,51 | 7.92 | 8,00 | 7,98 | 8,01 | 7,94 | 7,98 | 7,97 |
| Bottom Temp °C | 25,67 | 25,38 | 26.14 | 25.92 | 25,18 | 24,94 | 25,42 | 24.78 | 24,96 | 24.99 |
| Bottom Salinity | 30.49 | 34,26 | 24,46 | 24,17 | 31,98 | 33,71 | 31,98 | 34.53 | 34,42 | 34.38 |
| Bottom DO% | 106.7 | 109,5 | 74,1 | 105.6 | 102,0 | 101.2 | 102,8 | 87,8 | 95.8 | 93.3 |
| Bottom DO mg/L | 7,34 | 7,54 | 5.11 | 7,26 | 7,01 | 6,91 | 7,04 | 5.98 | 6,50 | 6,35 |
| Bottom pH | 8.00 | 8,00 | 7.72 | 8.02 | 7,99 | 7.98 | 7,99 | 7,94 | 7.98 | 7.97 |
| $NO_3^-$ (Nitrate) mg/L | 0,96 | 0.71 | 0.46 | 0.87 | 0.78 | 0.75 | 0,91 | 0.00 | 1,07 | 1,40 |
| $NO_2^-$ (Nitrite) mg/L | 0,000 | 0,000 | 0,000 | 0.000 | 0.000 | 0,000 | 0.005 | 0.000 | 0,000 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 | 0.00 | 0,00 | 0.02 | 0,00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.04 | 0.04 | 0.03 | 0.01 | 0.13 | 0.00 | 0.09 | 0.00 | 0.00 | 0.02 |
| $SiO_2$ (Silica) mg/L | 3 | 5 | 11 | 6 | 5 | 0 | 4 | 0 | 1 | 1 |
| Turbidity (FAU) | 3 | 4 | 21 | 9 | 5 | 0 | 10 | 1 | 0 | 1 |

# Water Quality Sampling Data Sheet:

Date: 5/[illegible]
Fishpond: [illegible]
Initial Observations: [illegible]

| | | | |
|---|---|---|---|
| **Start** Time | 1:00 pm | % Cloud Cover | 70% |
| **Finish** Time | | | |
| Tide | High | Precipitation | 0 |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | Orchard 5-10 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 57 | 62 | 60 | 63 | 70 | 64 | 55 | 59 | 78 | 67 |
| Surf. Water Temp °C | 27.39 | 26.47 | 25.59 | 27.51 | 27.94 | 27.36 | 27.56 | 28.05 | 26.73 | 27.36 |
| Surf. Salinity | 30.37 | 27.18 | 24.53 | 31.90 | 32.97 | 30.55 | 32.03 | 34.00 102.0 | 30.99 | 35.41 |
| Surf. DO % | 95.0 | 97.6 | 82.7 | 95.2 | 113.3 | 112.4 | 101.6 | 6.56 | 117.5 | 98.0 |
| Surf. DO mg/L | 6.53 | 6.72 | 5.72 | 6.30 | 7.39 | 7.59 | 6.75 | 7.89 | 7.67 | 6.42 |
| Surf. pH | 7.67 | 7.79 | 7.35 | 7.86 | 7.94 | 7.99 | 7.88 | 28.15 | 7.96 | 7.55 |
| Bottom Temp °C | 27.49 | 27.66 | 27.65 | 28.21 | 27.69 | 27.43 | 27.73 | 34.00 | 27.97 | 27.52 |
| Bottom Salinity | 30.49 | 33.60 | 32.37 | 32.93 | 33.31 | 34.56 | 32.29 | 103.6 | 35.00 | 33.62 |
| Bottom DO% | 94.0 | 108.4 | 93.0 | 104.5 | 113.3 | 129.8 | 106.8 | 6.74 | 112.2 | 99.5 |
| Bottom DO mg/L | 6.23 | 7.08 | 5.83 | 6.66 | 7.41 | 8.40 | 7.01 | 7.90 | 7.27 | 6.49 |
| Bottom pH | 7.72 | 7.80 | 7.85 | 7.87 | 7.93 | 8.00 | 7.90 | | 7.96 | 7.87 |
| $NO_3^-$ (Nitrate) mg/L | 0.90 | 0.94 | 1.06 | 1.03 | 1.32 | 0.55 | 0.90 | 1.84 | 1.70 | 0.87 |
| $NO_2^-$ (Nitrite) mg/L | 0.005 | 0.007 | 0.002 | 0.000 | 0.000 | 0.000 | 0.000 | 0.002 | 0.000 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.02 | 0.00 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.62 | 0.05 | 0.08 | 0.01 | 0.16 | 0.00 | — | 0.07 | 0.00 | 0.00 |
| $SiO_2$ (Silica) mg/L | 4 | 4 | 3 | 2 | 1 | 0 | 1 | 2 | 0 | 0 |
| Turbidity (FAU) | 5 | 5 | 7 | 2 | 0 | 5 | 6 | 0 | 0 | 4 |

# Water Quality Sampling Data Sheet:

Date: 6/4/15
Fishpond: Lelo
Initial Observations: Mowing

| | | | |
|---|---|---|---|
| **Start** Time | 2:00 pm | % Cloud Cover | 80% Overcast sunny earlier |
| **Finish** Time | 6/8/15 @ 10:00am | | |
| Tide | High | Precipitation | 0 |
| Air Temp | | Other Comments: | sample tested late, sorry busy trying to raise walls so they retain more water. |
| Wind Speed/Direction | Onshore 5 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 60 | 74 | 76 | 82 | 83 | 59 | 72 | 80 | 80 | 86 |
| Surf. Water Temp °C | 28.38 | 26.05 | 27.29 | 28.72 | 27,92 | 26.00 | 28.38 | 26.41 | 25.86 | 26,55 |
| Surf. Salinity | 31.05 | 1,51 | 18.86 | 26,42 | 28,70 | 33.08 | 30,21 | 32,31 | 37.79 | 32,31 |
| Surf. DO % | 105.6 | 105.5 | 101.3 | 134.9 | 142.1 | 148,8 | 149.9 | 128,6 | 116.8 | 123,5 |
| Surf. DO mg/L | 9,51 | 7.93 | 7.53 | 9,18 | 9,51 | 10,02 | 9,75 | 8,66 | 7,79 | 8,45 |
| Surf. pH | 7.72 | 7,31 | 7.84 | 8,01 | 7.98 | 7,85 | 8,03 | 7,91 | 7,86 | 7,92 |
| Bottom Temp °C | 28.34 | 27,31 | 28.55 | 24,63 | 27,71 | 25.98 | 27,25 | 26,47 | 25,86 | 26.77 |
| Bottom Salinity | 31,46 | 31.98 | 31.76 | 30,03 | 32,08 | 33.04 | 31.80 | 32.56 | 32.92 | 32,35 |
| Bottom DO% | 107,3 | 150.1 | 165.8 | 160.2 | 143,9 | 119,2 | 148,0 | 129,9 | 114,2 | 127,7 |
| Bottom DO mg/L | 9,13 | 10,02 | 10.74 | 10.16 | 9,54 | 8,03 | 9,53 | 8,67 | 7,75 | 8,52 |
| Bottom pH | 7,85 | 7,96 | 8,04 | 8,05 | 7,96 | 7,86 | 7,98 | 7,93 | 7,82 | 7,95 |
| $NO_3^-$ (Nitrate) mg/L | 0,10 | 0,08 | 0,00 | 0,20 | 0,23 | 0,00 | 1,14 | 0,00 | 1,04 | 0,00 |
| $NO_2^-$ (Nitrite) mg/L | 0,000 | 0,000 | 0,000 | 0,000 | 0,000 | 0,000 | 0,002 | 0,000 | 0,000 | 0,000 |
| $NH_4$ (Ammonia) mg/L | 0,00 | 0,03 | 0,05 | 0,02 | 0,07 | 0,04 | 0,00 | 0,00 | 0,00 | 0,00 |
| $PO_4^{-3}$ (Phosphate) | 0,03 | 0,10 | 0,03 | 0,02 | 0,00 | 0,20 | 0,02 | 0,00 | 0,00 | — |
| $SiO_2$ (Silica) mg/L | 8 | 14 | 12 | 4 | 4 | 0 | 2 | 2 | 0 | 0 |
| Turbidity (FAU) | 4 | 10 | 9 | 7 | 5 | 0 | 4 | 0 | 0 | 2 |

# Water Quality Sampling Data Sheet:

Date: 6/10/15
Fishpond: He'eia
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 11:43am | % Cloud Cover | 35% supersunny & hot |
| **Finish** Time | | | |
| Tide | high ↑ | Precipitation | 0 |
| Air Temp | 10mph | Other Comments: | |
| Wind Speed/Direction | onshore | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 58 | 69 | 55 | 60 | 77 | 62 | 56 | 59 | 58 | 65 | |
| Surf. Water Temp °C | 28.10 | 27.29 | 27.35 | 28.66 | 28.22 | 28.81 | 29.24 | 29.03 | 29.38 | 28.93 | 28.22 |
| Surf. Salinity | 25.33 | 19.28 | 10.61 | 27.30 | 28.20 | 33.82 | 27.90 | 33.19 | 34.42 | 29.33 | 28.20 |
| Surf. DO % | 123.3 | 111.5 | 96.9 | 119.5 | 123.6 | 133.3 | 130.5 | 130.3 | 121.8 | 117.9 | 123.6 |
| Surf. DO mg/L | 8.37 | 7.91 | 7.23 | 7.62 | 8.28 | 8.53 | 8.61 | 8.33 | 7.70 | 7.74 | 8.28 |
| Surf. pH | 7.82 | 7.74 | 7.42 | 7.78 | 7.80 | 7.83 | 7.88 | 7.83 | 7.85 | 7.81 | 7.84 |
| Bottom Temp °C | 28.79 | 28.40 | 29.00 | 28.81 | 28.39 | 28.88 | 29.33 | 29.08 | 29.43 | 29.32 | |
| Bottom Salinity | 25.75 | 31.03 | 26.00 | 28.82 | 27.79 | 33.81 | 27.99 | 33.21 | 34.43 | 31.34 | |
| Bottom DO% | 133.6 | 124.2 | 100.7 | 123.5 | 129.6 | 131.9 | 129.8 | 128.5 | 122.6 | 118.3 | |
| Bottom DO mg/L | 9.14 | 11.03 | 6.70 | 8.14 | 8.36 | 8.45 | 8.65 | 8.20 | 7.66 | 7.59 | |
| Bottom pH | 7.93 | 7.99 | 7.57 | 7.81 | 7.83 | 7.84 | 7.90 | 7.83 | 7.86 | 7.87 | |
| $NO_3^-$ (Nitrate) mg/L | 0.32 | 0.28 | 0.47 | 0.15 | 0.29 | 0.63 | 1.68 | 0.81 | 0.34 | 0.00 | |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.002 | 0.000 | 0.002 | 0.014 | 0.000 | 0.000 | 0.004 | — | |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.07 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 | 0.00 | |
| $PO_4^{-3}$ (Phosphate) | 0.27 | [illegible] | 0.13 | 0.04 | 0.09 | 0.05 | 0.06 | 0.80 | 0.09 | 0.06 | |
| $SiO_2$ (Silica) mg/L | 4 | 7 | 13 | 3 | 2 | 1 | 3 | 1 | 1 | 5 | |
| Turbidity (FAU) | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | 11 | 7 | |
| | 2 | 9 | 16 | 0 | 15 | 5 | 17 | 2 | 2 | 2 | |

# Water Quality Sampling Data Sheet:

Date: 6/10/15
Fishpond: Heʻeia
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 11:43am | % Cloud Cover | 35% Supersunny & hot |
| **Finish** Time | | | |
| Tide | high ↑ | Precipitation | 0 |
| Air Temp | 10mph | Other Comments: | |
| Wind Speed/Direction | onshore | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 58 | 69 | 55 | 60 | 77 | 62 | 56 | 59 | 58 | 65 | |
| Surf. Water Temp °C | 28.10 | 27.29 | 27.35 | 28.66 | 28.22 | 28.81 | 29.24 | 29.03 | 29.38 | 28.13 | 28.22 |
| Surf. Salinity | 25.33 | 19.28 | 10.61 | 27.30 | 28.20 | 33.82 | 27.90 | 33.19 | 34.42 | 29.33 | 28.20 |
| Surf. DO % | 123.3 | 111.5 | 96.9 | 114.5 | 123.6 | 133.3 | 130.5 | 130.3 | 121.8 | 117.9 | 123.6 |
| Surf. DO mg/L | 8.37 | 7.91 | 7.23 | 7.62 | 8.26 | 8.53 | 8.61 | 8.33 | 7.70 | 7.74 | 8.28 |
| Surf. pH | 7.82 | 7.74 | 7.42 | 7.78 | ~~28.30~~ 27.27 | 7.83 | 7.88 | 7.83 | 7.85 | 7.87 | 7.84 |
| Bottom Temp °C | 28.79 | 28.40 | 29.00 | 28.81 | ~~27.30~~ 27.84 | 28.88 | 29.33 | 29.08 | 29.43 | 29.32 | |
| Bottom Salinity | 25.75 | 31.03 | 26.00 | 28.82 | ~~31.04~~ 31.16 | 33.81 | 27.79 | 33.21 | 34.43 | 31.34 | |
| Bottom DO% | 133.6 | ~~129.~~ 130.2 | 100.7 | 123.5 | ~~127.16~~ 128.36 | 131.9 | 129.8 | 128.5 | 122.6 | 118.3 | |
| Bottom DO mg/L | 9.14 | 11.03 | 6.70 | 8.14 | 8.36 | 8.45 | 8.65 | 8.20 | 7.66 | 7.59 | |
| Bottom pH | 7.93 | 7.99 | 7.57 | 7.81 | 7.83 | 7.84 | 7.90 | 7.83 | 7.86 | 7.87 | |
| $NO_3^-$ (Nitrate) mg/L | 0.32 | 0.28 | 0.47 | 0.15 | 0.29 | 0.63 | 1.68 | 0.81 | 0.34 | 0.00 | |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.002 | 0.000 | 0.002 | 0.014 | 0.000 | 0.000 | 0.004 | — | |
| $NH_3$ (Ammonia) mg/L | 0.00 | 0.07 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 | 0.00 | |
| $PO_4^{-3}$ (Phosphate) | 0.27 | ~~0.10~~ | 0.13 | 0.04 | 0.09 | 0.05 | 0.06 | 0.80 | 0.09 | 0.06 | |
| $SiO_2$ (Silica) mg/L | 4 | ~~0.7~~ 7 | ~~3.13~~ | 3 | ~~2.2~~ 2 | ~~2.1~~ 1 | 3 | 1 | 1 | 5 | |
| Turbidity (FAU) | ~~20~~ | ~~15~~ | ~~16~~ | ~~0~~ | ~~25~~ | ~~5~~ | ~~17~~ | ~~2~~ | 11 | 7 | |
| | 2 | 9 | 16 | 0 | 15 | 5 | 17 | 2 | 2 | 2 | |

# Water Quality Sampling Data Sheet:

Date: 06/17/15
Fishpond: He'ei'a
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 12:52 pm | % Cloud Cover | 60% |
| **Finish** Time | | | |
| Tide | medium ↑ | Precipitation | light shower earlier |
| Air Temp | | Other Comments: | real bi-polar weather |
| Wind Speed/Direction | onshore 5-10 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 43 | 53 | 74 | 74 | 90 | 58 | 60 | 61 | 59 | 83 |
| Surf. Water Temp °C | 28.40 | 28.15 | 26.10 | 28.85 | 29.00 | 27,47 | 28,91 | 27,61 | 27,24 | 28.24 |
| Surf. Salinity | 28,91 | 29,13 | 17,84 | 28.10 | 28.46 | 34,15 | 29,64 | 33.23 | 34,11 | 33.19 |
| Surf. DO % | 94,3 | 130.1 | 84.2 | 107,3 | 123,2 | 98.0 | 104,8 | 106,7 | 98,2 | 104.6 |
| Surf. DO mg/L | 6,24 | 8,60 | 6,01 | 7,08 | 8,14 | 6,39 | 6.87 | 7,01 | 6,45 | 6,84 |
| Surf. pH | 7,75 | 7,96 | 7,57 | 7,86 | 7,92 | 7.86 | 7,84 | 7,87 | 7,84 | 7,90 |
| Bottom Temp °C | 28,42 | 28,09 | 28,94 | 28,81 | 28,86 | 27,48 | 27,92 | 27,59 | 27,26 | 28,55 |
| Bottom Salinity | 28.92 | 32,86 | 29,85 | 29,85 | 33,24 | 33,99 | 32,22 | 33.19 | 34,11 | 33,37 |
| Bottom DO% | 94,4 | 126,3 | 110 | 120.9 | 120.0 | 106,1 | 107,6 | 107.5 | 99,1 | 107,3 |
| Bottom DO mg/L | 6.15 | 8,22 | 7,95 | 7,98 | 7,69 | 6,86 | 7,05 | 7,06 | 6,99 | 6,91 |
| Bottom pH | 7,76 | 7,93 | 7.88 | 7,94 | 7,93 | 7,87 | 7.88 | 7,87 | 7.84 | 7,90 |
| $NO_3^-$ (Nitrate) mg/L | 0,67 | 1,22 | 0,68 | 0,71 | 0,73 | 1,27 | 1,30 | 1.23 | 0.75 | 1,04 |
| $NO_2^-$ (Nitrite) mg/L | 0,002 | 0,000 | 0.001 | 0,000 | 0,000 | 0,000 | 0.008 | 0.009 | 0.000 | 0,002 |
| $NH_1$ (Ammonia) mg/L | 0,02 | 0,03 | 0.04 | 0.03 | 0.00 | 0,00 | 0,07 | 0,03 | 0.01 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.04 | 0.05 | 0.18 | 0,07 | 0,27 | 0.05 | 0,13 | 0,11 | 0,25 | 0,05 |
| $SiO_2$ (Silica) mg/L | 5 | 4 | 12 | 4 | 4 | 0 | 3 | 2 | 0 | 1 |
| Turbidity (FAU) | 0 | 0 | 8 | 2 | 02 | 0 | 6 | 0 | 0 | 0 |

# Water Quality Sampling Data Sheet:

Date: 06/17/15
Fishpond: He'ei 9
Initial Observations: Maury

| | | | |
|---|---|---|---|
| **Start** Time | 12:52 pm | % Cloud Cover | 60% |
| **Finish** Time | | | |
| Tide | medium ↑ | Precipitation | light shower earlier |
| Air Temp | | Other Comments: | real bi-polar weather |
| Wind Speed/Direction | onshore 5-10 mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 43 | 53 | 74 | 74 | 90 | 58 | 60 | 61 | 59 | 83 |
| Surf. Water Temp °C | 28.40 | 28.15 | 26.10 | 28.85 | 29.00 | 27.47 | 28.91 | 27.61 | 27.24 | 28.24 |
| Surf. Salinity | 28.91 | 29.13 | 17.94 | 28.10 | 28.46 | 34.15 | 29.64 | 33.23 | 34.11 | 33.19 |
| Surf. DO % | 94.3 | 130.1 | 89.2 | 107.3 | 123.2 | 98.0 | 104.8 | 106.7 | 98.2 | 104.6 |
| Surf. DO mg/L | 6.24 | 8.60 | 6.01 | 7.08 | 8.14 | 6.39 | 6.87 | 7.01 | 6.45 | 6.89 |
| Surf. pH | 7.75 | 7.96 | 7.51 | 7.86 | 7.92 | 7.86 | 7.84 | 7.87 | 7.84 | 7.90 |
| Bottom Temp °C | 28.42 | 28.09 | 28.94 | 28.81 | 28.86 | 27.48 | 27.92 | 27.59 | 27.26 | 28.55 |
| Bottom Salinity | 28.92 | 32.86 | 29.85 | 29.85 | 33.24 | 33.99 | 32.22 | 33.19 | 34.11 | 33.37 |
| Bottom DO% | 94.4 | 126.3 | 77.0 | 102.9 | 120.0 | 106.1 | 107.6 | 107.5 | 99.1 | 107.3 |
| Bottom DO mg/L | 6.15 | 8.22 | 7.95 | 7.98 | 7.69 | 6.86 | 7.05 | 7.06 | 6.99 | 6.91 |
| Bottom pH | 7.76 | 7.93 | 7.88 | 7.94 | 7.93 | 7.87 | 7.88 | 7.87 | 7.84 | 7.90 |
| $NO_3^-$ (Nitrate) mg/L | 0.67 | 1.22 | 0.68 | 0.71 | 0.73 | 1.27 | 1.30 | 1.23 | 0.75 | 1.04 |
| $NO_2^-$ (Nitrite) mg/L | 0.002 | 0.000 | 0.001 | 0.000 | 0.000 | 0.000 | 0.008 | 0.009 | 0.000 | 0.002 |
| $NH_4$ (Ammonia) mg/L | 0.02 | 0.03 | 0.04 | 0.03 | 0.00 | 0.00 | 0.07 | 0.03 | 0.01 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.04 | 0.05 | 0.18 | 0.07 | 0.27 | 0.05 | 0.13 | 0.11 | 0.25 | 0.05 |
| $SiO_2$ (Silica) mg/L | 5 | 4 | 12 | 4 | 4 | 0 | 3 | 2 | 0 | 1 |
| Turbidity (FAU) | 0 | 0 | 8 | 2 | 2 | 0 | 6 | 0 | 0 | 0 |

# Water Quality Sampling Data Sheet:

Date: 06/23/15
Fishpond: He'eia
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | | % Cloud Cover | 30% Sunny |
| **Finish** Time | | | |
| Tide | high ↓ | Precipitation | rained in morning → super sunny |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | Onshore mod steadily | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 48 | 58 | 54 | 61 | 68 | 49 | 52 | 52 | 60 | 58 |
| Surf. Water Temp °C | 25.79 | 25.29 | 25.25 | 26.17 | 26.07 | 26.22 | 26.65 | 26.60 | 26.96 | 26.77 |
| Surf. Salinity | 25.07 | 21.9 | 10.05 | 25.15 | 24.85 | 33.75 | 26.37 | 32.19 | 34.16 | 29.06 |
| Surf. DO % | 89.5 | 91.4 | 83.4 | 93.1 | 87.0 | 99.6 | 92.6 | 99.9 | 103.2 | 93.7 |
| Surf. DO mg/L | 6.35 | 6.65 | 6.41 | 6.54 | 6.12 | 6.65 | 6.37 | 6.69 | 6.79 | 6.35 |
| Surf. pH | 7.70 | 7.71 | 7.51 | 7.76 | 7.74 | 7.79 | 7.78 | 7.81 | 7.87 | 7.83 |
| Bottom Temp °C | 26.51 | 26.35 | 27.64 | 26.35 | 26.59 | 26.30 | 26.70 | 26.63 | 26.97 | 27.11 |
| Bottom Salinity | 25.96 | 26.60 | 22.79 | 27.83 | 27.30 | 33.76 | 26.36 | 32.17 | 34.16 | 29.60 |
| Bottom DO% | 91.9 | ~~94.1 99.6~~ | 72.3 | 94.4 | 80.7 | 99.9 | 91.8 | 99.8 | 103.1 | 92.3 |
| Bottom DO mg/L | 6.40 | 7.14 | 5.06 | 6.51 | 5.56 | 6.68 | 6.33 | 6.68 | 6.97 | 6.56 |
| Bottom pH | 7.76 | 7.85 | 7.59 | 7.78 | 7.75 | 7.79 | 7.78 | 7.81 | 7.87 | 7.85 |
| $NO_3^-$ (Nitrate) mg/L | 1.05 | 1.01 | 0.57 | 1.00 | 0.00 | 0.19 | 0.00 | 0.59 | 0.00 | 0.00 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.006 | 0.002 | 0.000 | 0.001 | 0.000 | 0.005 | 0.003 | 0.004 |
| $NH_4$ (Ammonia) mg/L | 0 | 0 | 5 | 0 | 0 | 0 | 0 | 0 | — | 0 |
| $PO_4^{-3}$ (Phosphate) | 0.08 | 0.14 | 0.12 | 0.12 | 0.08 | 0.00 | 0.05 | 0.03 | 0.01 | 0.00 |
| $SiO_2$ (Silica) mg/L | 0.00 | 0.00 | 0.00 | 0.03 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Turbidity (FAU) | 5 | 0 | 25 | ~~3~~ 3 | 2 | 0 | 0 | 0 | 0 | 0 |

#3
54
25 25
10 05
93 4
6.41
7.51
27.64
~~22.79~~
~~72.3~~
5.06
7.59

# Water Quality Sampling Data Sheet:

Date: 6/25/16
Fishpond: Heeia
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 12:45 | % Cloud Cover | 40% Sunny |
| **Finish** Time | | | |
| Tide | high ↑ | Precipitation | small rain in morning, sunny now |
| Air Temp | | Other Comments: | small squal ~~and~~ mediumrain, ~10min. Then overcast w/ moments of sunny. |
| Wind Speed/Direction | onshore 5mph-8 | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 55 | 68 | 66 | 71 | 92 | 66 | 61 | 64 | 71 | 72 |
| Surf. Water Temp °C | 26.43 | 26.32 | 26.11 | 27.25 | 27.41 | 27.24 | 27.71 | 27.53 | 27.26 | 26.85 |
| Surf. Salinity | 24.59 | 17.87 | 21.04 | 24.03 | 25.13 | 32.28 | 27.56 | 32.6 | 34.36 | 27.27 |
| Surf. DO % | 113.3 | 105.7 | 99.1 | 105.2 | 110.8 | 106.6 | 102.2 | 116 | 108.5 | 103.2 |
| Surf. DO mg/L | 7.95 | 7.70 | 7.15 | 7.31 | 7.49 | 7.20 | 6.89 | 7.67 | 7.17 | 7.09 |
| Surf. pH | 7.83 | 7.78 | 7.86 | 7.88 | 7.91 | 7.88 | 7.89 | 7.94 | 7.96 | 7.95 |
| Bottom Temp °C | 27.35 | 27.23 | 28.25 | 27.83 | 27.42 | 27.32 | 28.25 | 27.58 | 27.57 | 27.91 |
| Bottom Salinity | 27.41 | 31.43 | 27.23 | 27.58 | 26.67 | 32.40 | 31.14 | ~~32.57~~ 27.52 | 34.3 | 32.09 |
| Bottom DO% | 115.3 | 107.6 | 76.2 | 102.8 | 110.2 | 106.5 | 119.7 | ~~121.6~~ 92.5 | 110.7 | 123.5 |
| Bottom DO mg/L | 7.33 | 7.12 | 5.05 | 6.94 | 7.44 | 7.04 | 7.82 | 7.92 | 7.23 | 8.10 |
| Bottom pH | 7.86 | 7.85 | 7.73 | 7.88 | 7.92 | 7.90 | 7.9 | 7.95 | 7.98 | 7.97 |
| $NO_3^-$ (Nitrate) mg/L | 0.95 | 0.72 | 0.78 | 0.76 | 0.74 | 0.92 | 0.88 | 1.03 | 0.99 | 0.59 |
| $NO_2^-$ (Nitrite) mg/L | 0.005 | 0.001 | 0.004 | 0.003 | 0.000 | 0.000 | 0.002 | 0.000 | 0.000 | 0.002 |
| ~~$NH_4$ (Ammonia) mg/L~~ | 1 | 6 | 13 | 2 | 3 | 0 | 11 | 0 | 0 | 2 |
| $PO_4^{-3}$ (Phosphate) | 0.04 | 0.05 | 0.17 | 0.04 | 0.08 | 0.08 | 0.11 | 0.04 | 0.00 | 0.09 |
| $SiO_2$ (Silica) mg/L | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Turbidity (FAU) | 0.00 | 0.00 | 0.05 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 |

Turbidity

wrote in wrong slot sorry

Ammonia

# Water Quality Sampling Data Sheet:

Date: 7/1/15
Fishpond: He'eia
Initial Observations: Maury

| | | | |
|---|---|---|---|
| **Start** Time | 1:43pm | % Cloud Cover | 10% all over mountain more in bay |
| **Finish** Time | | | |
| Tide | high ↑ | Precipitation | 0 |
| Air Temp | | Other Comments: | |
| Wind Speed/Direction | onshore easterly | | |

HOT!!

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 72 | 82 | 97 | 95 | 101 | 89 | 97 | 91 | 102 | 102 |
| Surf. Water Temp °C | 34,97 | 32,66 | 33.45 | 35,57 | 35,01 | 29,36 | 35,43 | 30,87 | 30,06 | 32,05 |
| Surf. Salinity | 32,09 | 33,75 | 32,34 | 31,46 | 33,05 | 34,75 | 32,93 | 34,60 | 34,63 | 33,61 |
| Surf. DO % | 165,8 | 150,4 | 106,3 | 141,9 | 136,8 | 107,3 | 160,8 | 116,2 | 117,9 | 119,9 |
| Surf. DO mg/L | 9,83 | 9,11 | 6,79 | 8,78 | 8,22 | 6,80 | 9,59 | 7,16 | 7,36 | 7,30 |
| Surf. pH | 7,99 | 7,99 | 7,83 | 7,94 | 6,01 | 7,91 | 7,96 | 7,92 | 7,93 | 7,92 |
| Bottom Temp °C | 34,75 | 30,26 | 33,64 | 32,91 | 30,25 | 29,32 | 31,22 | 30,87 | 30,03 | 31,43 |
| Bottom Salinity | 33,72 | 34,69 | 33,95 | 34,32 | 34,20 | 34,74 | 34,47 | 34,60 | 34,65 | 34,37 |
| Bottom DO% | 169,3 | 123,5 | 134,9 | 132,9 | 119,1 | 107,4 | 126,4 | 116,7 | 114,6 | 123,6 |
| Bottom DO mg/L | 9,22 | 7,63 | 8,02 | 7,88 | 7,31 | 6,78 | 7,74 | 7,19 | 7,15 | 7,56 |
| Bottom pH | 7,98 | 7,95 | 7,93 | 7,94 | 7,96 | 7,91 | 7,94 | 7,92 | 7,93 | 7,97 |
| $NO_3^-$ (Nitrate) mg/L | 1,23 | 0,67 | 0,97 | 1,45 | 1,19 | 0,86 | 0,71 | 1,15 | 0,81 | 1,27 |
| $NO_2^-$ (Nitrite) mg/L | 0,000 | 0,000 | 0,004 | 0,005 | 0,002 | 0,001 | 0,000 | 0,000 | 0,000 | 0,000 |
| $NH_4$ (Ammonia) mg/L | 0,00 | 0,00 | 0,00 | 0,00 | 0,00 | 0,00 | 0,02 | 0,00 | 0,00 | 0,00 |
| $PO_4^{-3}$ (Phosphate) | 0,00 | 0,06 | 0,12 | 0,10 | 0,04 | 0,03 | 0,00 | 0,04 | 0,17 | 0,11 |
| $SiO_2$ (Silica) mg/L | 4 | 6 | 4 | 5 | 4 | 2 | 2 | 2 | 1 | 3 |
| Turbidity (FAU) | 0 | 2 | 7 | 7 | 3 | 0 | 4 | 0 | 0 | 0 |

#3
97
33.45
29.34
106.3
6.79
7.83
33.64
33.95
134.9
8.02
7.93

# Water Quality Sampling Data Sheet:

Date: 7/7/15
Fishpond: Heeia Kelii

Initial Observations:

| | | | |
|---|---|---|---|
| **Start** Time | 3:45p 7/7 | % Cloud Cover | 95% cool breeze but warm temps |
| **Finish** Time | | | |
| Tide | low rising | Precipitation | ∅ |
| Air Temp | at least 88° | Other Comments: | |
| Wind Speed/Direction | ENE 10mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 45 | 52 | 54 | 54 | 81 | 52 | 50 | 52 | 55 | 62 |
| Surf. Water Temp °C | 31.62 | 31.15 | 28.44 | 31.30 | 30.87 | 30.24 | 31.51 | 31.65 | 29.46 | 31.09 |
| Surf. Salinity | 30.57 | 27.92 | 14.70 | 27.47 | 27.32 | 34.96 | 29.17 | 30.90 | 34.98 | 31.04 |
| Surf. DO % | 191.1 | 208.9 | 111.3 | [illegible] | 155.1 | 135.6 | 173.7 | 182.2 | 127.9 | 133.1 |
| Surf. DO mg/L | 11.53 | 13.29 | 7.88 | 10.24 | 9.96 | 8.40 | 10.96 | 11.28 | 8.05 | 8.40 |
| Surf. pH | 7.96 | 8.34 | [illegible] | 8.22 | 8.19 | 8.07 | 8.23 | 8.41 | 8.06 | 8.18 |
| Bottom Temp °C | 31.74 | 31.17 | 31.17 | 31.50 | 31.86 | 30.24 | 31.57 | 31.65 | 29.46 | 31.12 |
| Bottom Salinity | 30.60 | 27.82 | 25.71 | 28.90 | 30.66 | 34.95 | 29.21 | 30.89 | 34.98 | 31.13 |
| Bottom DO% | 178.3 | 208.3 | 158.3 | 175.6 | 164.6 | 128.8 | 178.6 | 180.3 | 126.7 | 131.2 |
| Bottom DO mg/L | 11.07 | 13.27 | 9.94 | 11.09 | 10.32 | 8.09 | 11.22 | 11.30 | 7.95 | 8.19 |
| Bottom pH | 8.10 | 8.34 | 8.03 | 8.23 | 8.23 | 8.06 | 8.25 | 8.41 | 8.04 | 8.18 |
| $NO_3^-$ (Nitrate) mg/L | 0.42 | 0.00 | 0.00 | 0.00 | 0.00 | 0.23 | 0.00 | 0.00 | 0.20 | 0.00 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.003 | 0.005 | 0.000 | 0.004 | 0.000 | 0.000 | 0.002 | 0.000 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0.00 | 0.00 | 0.02 | 0.00 | 0.00 | 0.00 | 0.02 | 0.00 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.00 | 0.00 | 0.00 | 0.10 | 0.08 | 0.00 | — | 0.03 | 0.02 | 0.07 |
| $SiO_2$ (Silica) mg/L | 4 | 4 | 4 | 6 | 6 | 2 | 3 | 2 | 1 | 4 |
| Turbidity (FAU) | 2 | 5 | 17 | 12 | 5 | 0 | 7 | 4 | 3 | 2 |

# Water Quality Sampling Data Sheet:

Date: 7/23/15
Fishpond: Heeia
Initial Observations: Mauka

| | | | |
|---|---|---|---|
| **Start** Time | 10:54am | % Cloud Cover | 40% |
| **Finish** Time | | | |
| rise ↑ ebb → Tide | high ↑E↓ | Precipitation | 0 |
| Air Temp | not sun w/ cool winds | Other Comments: | |
| Wind Speed/Direction | Onshore 5mph | | |

| Site # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Depth (cm) | 39 | 50 | 49 | 54 | 57 | 49 | 43 | 45 | 65 | 57 |
| Surf. Water Temp °C | 28.38 | 26.64 | 26.07 | 27.25 | 27.02 | 27.58 | 28.29 | 27.71 | 27.91 | 27.83 |
| Surf. Salinity | 29.69 | 22.58 | 7.82 | 20.50 | 16.88 | 33.57 | 24.51 | 33.71 | 34.11 | 37.63 |
| Surf. DO % | 100.7 | 97.4 | 75.9 | 91.5 | 90.5 | 103.7 | 116.9 | 118.7 | 98.0 | 1023 |
| Surf. DO mg/L | 6.81 | 6.58 | 5.80 | 9.01 | 6.48 | 7.73 | 7.98 | 7.74 | 6.36 | 6.51 |
| Surf. pH | 7.84 | 7.8 | 7.51 | 7.88 | 7.87 | 7.88 | 7.98 | 7.94 | 7.91 | 7.93 |
| Bottom Temp °C | 28.17 | 26.73 | 28.79 | 27.64 | 28.33 | 27.61 | 28.19 | 27.72 | 28.02 | 28.00 |
| Bottom Salinity | 26.98 | 25.55 | 24.16 | 29.64 | 26.51 | 33.55 | 29.01 | 33.68 | 34.13 | 32.53 |
| Bottom DO% | 97.5 | 101.2 | 70.2 | 104.9 | 115.9 | 106.5 | 113.0 | 115.8 | 98.0 | 106.6 |
| Bottom DO mg/L | 6.55 | 7.03 | 4.79 | 7.23 | 7.79 | 6.97 | 7.68 | 7.75 | 6.32 | 6.99 |
| Bottom pH | 7.84 | 7.92 | 7.59 | 7.92 | 8.05 | 7.90 | 7.99 | 7.95 | 7.92 | 7.96 |
| $NO_3^-$ (Nitrate) mg/L | 0.47 | 0.00 | 0.00 | 0.45 | 0.46 | 0.00 | 0.67 | 0.00 | 0.00 | 0.30 |
| $NO_2^-$ (Nitrite) mg/L | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| $NH_4$ (Ammonia) mg/L | 0.01 | 0.00 | 0.05 | 0.00 | 0.00 | 0.00 | 0.00 | 0.03 | 0.00 | 0.00 |
| $PO_4^{-3}$ (Phosphate) | 0.00 | 0.11 | 0.07 | 0.07 | 0.11 | — | 0.09 | — | 0.08 | 0.12 |
| $SiO_2$ (Silica) mg/L | 6 | 7 | 16 | 6 | 9 | 0 | 5 | 0 | 0 | 2 |
| Turbidity (FAU) | 0 | 0 | 4 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |